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Respirator Surge Manufacturing

Options to Ensure U.S. Small and Medium-Sized
Enterprises Can Surge the Manufacturing of Respiratory
Personal Protective Equipment

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About This Report

Shortages of personal protective equipment (PPE) during the coronavirus disease 2019 (COVID-19) pandemic, particularly N95 respirators and elastomeric half-mask respirators, prompted the United States to seek additional sources of supply beyond traditional large and foreign suppliers. In response, a range of small and medium-sized enterprises (SMEs) entered or expanded into respirator manufacturing but faced significant challenges in scaling production, accessing stable demand and markets, and navigating regulatory requirements. This report builds on earlier research on SME surge manufacturing during COVID-19, observing that these challenges persist across the supply, demand, and regulatory domains. It provides targeted recommendations to strengthen SME resilience and competitiveness through reforms in policy, stockpiling practices, and market incentives. The analysis is intended for policymakers, industry leaders, and researchers seeking to support a more robust and adaptable domestic PPE manufacturing base for future biological emergencies.

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Summary

Issue

The United States still lacks a resilient domestic manufacturing base that can rapidly manufacture respiratory personal protective equipment (PPE) during biological emergencies. Although the coronavirus disease 2019 (COVID-19) pandemic exposed dangerous shortages of N95 respirators and other protective equipment, many of the structural barriers that constrained domestic production remain unresolved. Small and medium-sized enterprises (SMEs) represent an important but underleveraged part of a more resilient domestic industrial base: Their flexibility, regional distribution, and potential for surge production make them well positioned to support emergency response, but current market and policy conditions leave these SMEs at a persistent disadvantage.

U.S. PPE SMEs face challenges across the full production life cycle, including access to raw materials, capital for specialized equipment, regulatory approval pathways, and procurement systems that favor large incumbent firms and group purchasing organizations (GPOs). At the same time, continued dependence on foreign suppliers, especially China, leaves the United States vulnerable to export restrictions, geopolitical instability, and global demand shocks that can cut off access to essential PPE at the moment of greatest need.

This is not simply a manufacturing issue; it is a matter of public health preparedness and national security. Respiratory PPE is a critical line of defense in biological crises, and the inability to produce adequate volumes of certified, high-quality equipment domestically weakens the nation's emergency response capacity. Foreign dependence and concentrated supply chains put health care and critical infrastructure workers at risk, limit surge responsiveness, and undermine confidence in government preparedness.

Against this backdrop, the key policy question is not whether the United States needs stronger domestic PPE capacity but how to build and sustain it, which can also set precedents for broader supply chain resilience. In this report, we examine the most consequential barriers U.S. PPE SMEs face, assess whether those challenges have changed since COVID-19, and evaluate the effectiveness of current policy tools, including domestic sourcing requirements, certification processes, and stockpiling strategies. We also identify where gaps in implementation, weak enforcement, and unstable demand continue to limit SME participation and explore what reforms to procurement, regulation, and market-shaping policies could better position U.S. firms to scale production before and during the next biological crisis.

The ability of U.S. SMEs to rapidly surge-manufacture respiratory PPE is essential to national pandemic preparedness and broader resilience against catastrophic biological threats. Without a strong domestic industrial base that can scale production quickly and operate independently of fragile global supply chains, the United States remains vulnerable to the same failures exposed during the COVID-19 pandemic. Supply disruptions, foreign export restrictions, and spikes in global demand left health care workers and the public without adequate protection when it was needed most. Strengthening SME surge capacity is therefore not only an industrial policy priority but a core requirement for public health security and emergency preparedness.

Approach

This research built on a baseline analysis of the Carnegie Mellon University examination of barriers affecting PPE SMEs during COVID-19. The report addresses two gaps in the university's scope—its exclusion of elastomeric half-mask respirators and the limitation of analysis to the 2020–2022 time frame—to guide further inquiry into whether those barriers persist today and what new challenges may have emerged. To

address these gaps, we conducted a literature review drawing on open-source media, peer-reviewed articles, and grey literature (government and technical reports, company websites) using such platforms as OpenAlex, ProQuest, Google Search, and large language models, focusing searches on PPE, mask manufacturing, COVID-19 supply chains, and U.S. manufacturing.

We complemented the literature review with semistructured interviews conducted between December 1, 2025, and April 3, 2026, with subject-matter experts spanning mask development and manufacturing, Occupational Safety and Health Administration (OSHA) and National Institute for Occupational Safety and Health (NIOSH) standards, emergency operations, domestic manufacturing policy, and the Strategic National Stockpile. Key topics included experiences in respiratory PPE manufacturing before, during, and after COVID-19; the effectiveness of existing domestic production policies; feedback on previously identified market entry barriers; regulatory approval processes; and opportunities to better support domestic SMEs.

Key Findings

Our analysis identified key limitations across three areas: (1) supply, production, and capacity challenges; (2) demand and market access constraints; and (3) regulatory hurdles. The following findings highlight the critical issues within each of these areas:

- Supply, production, and capacity
 - **High capital and equipment costs remain a major barrier to SME participation in respiratory PPE manufacturing.** SMEs face steep up-front costs for specialized machinery and automation, which are often necessary to achieve the scale and per-unit efficiency needed to compete with low-cost foreign producers. These cost pressures continue to weaken domestic production capacity and make it difficult for smaller firms to enter, sustain themselves in, or expand in the market.
 - **Workforce instability is both a symptom and a driver of SME fragility in the PPE sector.** SMEs struggle to retain skilled workers because inconsistent demand, lower compensation, and a shrinking manufacturing talent pipeline make long-term workforce planning difficult. These workforce challenges further reduce the ability of domestic firms to maintain production readiness and respond quickly when demand surges.
- Demand and market access
 - **Constrained market access is a structural obstacle that limits SME participation even when firms can produce quality PPE.** GPOs, distributors, and incumbent supplier relationships continue to dominate health care procurement and often exclude smaller domestic manufacturers from meaningful contracting opportunities. As a result, procurement channels remain concentrated among large established vendors, reducing supplier diversity and reinforcing supply chain fragility.
 - **Unstable demand is the single most important factor undermining SME survival and long-term readiness.** Many SMEs demonstrated that they could produce compliant respiratory PPE during COVID-19 but could not sustain operations once crisis demand receded. Without predictable demand between emergencies, firms cannot justify capital investment, maintain workforce capacity, or preserve surge readiness, leading many to exit the market entirely.
 - **Current stockpiling strategies are necessary but insufficient on their own to sustain domestic manufacturing capacity.** Heavy reliance on federal stockpiles does not adequately support a living manufacturing base that can remain operational and surge during a crisis. Stockpiling remains an essential preparedness tool but cannot, on its own, substitute for a stable, exercised, and commercially viable domestic PPE manufacturing ecosystem.

- Regulatory
 - **Certification and compliance requirements are manageable in principle but disproportionately burdensome for smaller manufacturers in practice.** SMEs often lack the in-house regulatory expertise, standardized procedures, and compliance resources available to larger firms. Combined with instability in the certification environment and weak enforcement against counterfeit or ineffective products, this creates avoidable barriers to entry, slows market participation, and undermines trust in lesser-known domestic suppliers.
 - **Existing domestic sourcing laws are weakened by loopholes, inconsistent definitions, and limited accountability.** Although such statutes as the Make PPE in America Act, Berry Amendment, Kissell Amendment, and Buy American Act were designed to support domestic manufacturing, agencies can still rely on waivers, narrow price comparisons, weak nonavailability determinations, and ambiguous definitions of the term *domestically manufactured* to continue purchasing offshore products. This undermines the intent of domestic sourcing policy and reduces its practical effect on market demand for U.S.-made PPE.

Recommendations

To strengthen U.S. preparedness for future biological crises, policymakers and other stakeholders should pursue targeted actions that improve the resilience and competitiveness of domestic small and medium-sized respiratory PPE manufacturers. A central finding of this analysis is that supporting PPE SMEs and strengthening domestic manufacturing capacity are functionally equivalent objectives: Because SMEs represent the predominant class of domestic producers, policies that improve their viability directly advance national supply chain resilience. Our recommendations identify practical steps to address barriers related to supply and production, demand and market access, and regulatory compliance but do not include a cost analysis. Although many recommendations address challenges across multiple areas, we have organized them into these three distinct categories, along with a fourth for overarching recommendations:

- Overall recommendations for strengthening SMEs and domestic PPE manufacturing
 - Congress and federal policymakers should consider **prioritizing a coordinated strategy to strengthen domestic respiratory PPE manufacturing through U.S. SMEs.** SME resilience is closely tied to national preparedness, supply chain security, and domestic industrial capacity.
 - The broader preparedness and policy community should **integrate the goal of supporting PPE SMEs and the goal of supporting domestic manufacturing.** In this sector, the two goals are largely aligned. Policies that sustain SME participation can also reduce dependence on foreign suppliers, diversify the industrial base, and improve the nation’s ability to respond rapidly during future biological crises.
- Overcoming supply, production, and capacity challenges
 - Congress, the U.S. Small Business Administration, and relevant federal agencies should consider **addressing supply, production, and capacity constraints** by expanding affordable equipment financing; supporting warm-basing¹ arrangements and precrisis production readiness; funding tech-

¹ Warm-basing is an approach to maintain standby manufacturing capacity.

nical assistance for automation and scale-up; and creating incentives to help SMEs recruit and retain skilled manufacturing, quality assurance, and regulatory personnel.

- Overcoming demand and market access constraints
 - The U.S. Department of Health and Human Services, the U.S. Department of Homeland Security, the U.S. Department of War,² the U.S. Department of Veterans Affairs, the Centers for Medicaid and Medicare Services (CMS), the Administration for Strategic Preparedness and Response, and state governments should **address demand and market access barriers by creating more-stable and more-predictable demand for domestic PPE**. Options include using long-term contracts, reforming federal and nonfederal stockpiling strategies, requiring hospitals receiving federal reimbursement to source a portion of PPE from SMEs, and encouraging state or regional stockpiling models that include domestic SME suppliers.
 - Congress, the U.S. Trade Representative, the U.S. Department of Commerce, the U.S. Department of Health and Human Services, the Internal Revenue Service, CMS, and state legislatures should consider **improving the price competitiveness of domestic SMEs by considering targeted tax incentives for health care purchasers that buy from U.S. SME manufacturers** and, when appropriate, evaluating tariffs or related trade tools to reduce the pricing advantage of foreign-made PPE.
 - CMS, the Federal Trade Commission, the U.S. Department of Justice, Congress, the U.S. Department of Veterans Affairs, and other oversight bodies should consider **promoting market fairness by addressing anticompetitive procurement practices that disadvantage SMEs**, particularly within GPOs and distributor contracting. This could include banning exclusionary or bundling practices, increasing transparency in vendor selection and fees, and requiring procurement models that preserve opportunities for qualified SME suppliers.
- Overcoming regulatory hurdles
 - Congress, the Federal Acquisition Regulation Council, the U.S. Government Accountability Office, agency inspectors general, and the Made in America Office should consider **reducing regulatory and procurement loopholes** by tightening waiver criteria for domestic sourcing, requiring total cost of ownership analysis instead of simple unit-price comparisons, establishing a clear and uniform definition of what constitutes *domestically manufactured* PPE, and publicly reporting waiver use and nondomestic procurement decisions.
 - NIOSH, the U.S. Food and Drug Administration, and SMEs themselves should **improve regulatory compliance capacity** by publishing clearer certification guidance, standardized checklists, and validation benchmarks, while also encouraging SMEs to form collaborative networks to share compliance tools, quality-control practices, and standard operating procedures. These actions should be considered alongside acknowledging and addressing the funding and resource constraints NIOSH may have in improving the approval process.

The central challenge is not whether domestic SMEs can produce respiratory PPE but whether U.S. policy creates the conditions for them to remain viable, competitive, and surge ready. Taken together, the findings show that the barriers SMEs face are interconnected across production, regulation, procurement, workforce, and demand. Although supply chain constraints and regulatory burdens present persistent challenges for SME manufacturers, demand instability remains the central threat to long-term viability; without predictable procurement, structural investments in domestic capacity are difficult to sustain. The recommendations in this report therefore point to a coordinated policy approach:

² The Department of War is designated the Department of Defense under Public Law 81-216, National Security Act Amendments of 1949.

lowering supply-side barriers, stabilizing demand, reforming procurement practices, closing domestic sourcing loopholes, and improving regulatory clarity. Without sustained action across these areas, the United States risks entering the next biological emergency with the same supply chain vulnerabilities exposed during COVID-19.

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Background

Introduction

Interventions against infectious diseases can take many forms, from medical countermeasures to environmental controls. Physical barriers that limit exposure to infectious microorganisms play a central role within this broader toolkit. Personal protective equipment (PPE) functions as a physical barrier to prevent ingestion of, inhalation of, and exposure to pathogens. Face masks, gowns, gloves, eye and face protection, and shoe protection help reduce the risk of exposure and subsequent disease. These resources are vital to protect both essential workers and the public against infectious diseases.

Although health care is a major user of PPE, other critical sectors rely on PPE as part of normal operations. Data from O*NET provide estimates by occupation of which occupations require PPE in the form of respirators and protective gowns.¹ This is combined with data from the Bureau of Labor Statistics Occupation Employment Statistics to estimate the number of workers in critical infrastructure sectors that require PPE.² This provides an estimate that the baseline demand for PPE in the United States is approximately 6.7 million workers, of which about two-thirds are in the health care sector. The list in Appendix E provides a breakdown by sector of PPE demand.

Despite PPE's critical role in reducing respiratory disease transmission, the United States and other countries faced severe supply shortages and lacked the capacity to manufacture it at sufficient scale and speed during the coronavirus disease 2019 (COVID-19) pandemic. The crisis triggered widespread disruptions affecting PPE supply, including high morbidity and mortality, international border closures, social and economic shutdowns, mask mandates, and escalating geopolitical tensions, while exposing structural vulnerabilities of hyperglobalized supply chains. Limited contemporary experience with pandemic-scale crises compounded these pressures, producing cascading disruptions that severely destabilized established PPE supply networks.

To address these PPE shortages, U.S. institutions—long reliant on a small number of large suppliers—pursued alternative sources, including small and medium-sized enterprises (SMEs) that emerged during the pandemic. Although the public health crisis generated significant demand for PPE, SMEs faced severe limitations in entering the market.

Existing research on the barriers PPE SMEs could face in future biological crises remains limited. A 2023 Carnegie Mellon University (CMU) study provided an important early assessment of SME challenges during COVID-19 but focused on a narrow range of PPE and only the 2020–2022 time frame.³ Important gaps remain in understanding how these constraints have persisted or evolved and how newer supply chain, market, or policy dynamics are shaping the current U.S. PPE SME landscape.

¹ O*Net, webpage, last updated May 19, 2026.

² U.S. Bureau of Labor Statistics, "Occupational Employment and Wage Statistics," webpage, undated.

³ Nikhil Kalathil, Granger M. Morgan, and Erica R. H. Fuchs, "Short-Term Economic Dynamism as a Policy Tool to Address Supply Shortages During Crises," *Industrial and Corporate Change*, Vol. 33, No. 4, August 2024.

Glossary

domestic manufacturer	A PPE manufacturer that is based in the United States and performs, at a minimum, final manufacturing processes
meltblown	A fabric created using high-velocity air to blow molten polypropylene, which creates fine fibers with a random arrangement ^a
PPE manufacturer	An entity that exclusively, or, as part of its operations, manufactures PPE
PPE supply chain	The entire process by which PPE is manufactured and distributed
SME	Small to medium-sized enterprises (less than \$50 million in annual sales) that manufacture PPE
spunbond	A fabric created by extruding continuous filaments that are laid down on a conveyor belt ^b
surge manufacturing	PPE manufacturing performed under conditions of increased demand and resulting in increased output
warm-basing	An approach to maintain standby manufacturing capacity

^a Purvis King, "Spunbond vs Meltblown: Understanding the Differences," *Palmetto Industries* blog, July 2, 2024.

^b King, 2024.

In this report, we analyze the demand-side, supply-side, and regulatory obstacles SMEs faced during COVID-19 in scaling PPE production and provide recommendations to strengthen U.S. PPE SME resilience, competitiveness, and surge-manufacturing capacity in preparation for future biological crises. Because SMEs represented the majority of domestic PPE producers that scaled up during COVID-19, this report treats investment in SME viability and investment in domestic manufacturing capacity as two sides of the same coin. The analysis addresses gaps in existing research by extending beyond the initial COVID-19 period and examining a broader range of respirators, including both N95 respirators and elastomeric half-mask respirators (EHMRs). Although many of the challenges that we discuss in this report are common across the broader PPE manufacturing landscape, the focus here is specifically on respiratory PPE. Accordingly, all references to *PPE* throughout this document should be understood to mean respiratory PPE unless otherwise stated. Although PPE is used broadly, the highest use, and thus demand, for PPE comes from health care, which significantly shaped the findings and recommendations of this report. The systemic issues, such as inconsistent demand and loopholes, affect all users of PPE, especially during public health emergencies. The research identifies specific barriers to market entry and long-term participation, rooted in the broader policy environment and political economy of domestic PPE procurement and production. These dynamics, which include structural supply chain constraints, concentrated purchasing power, and regulatory complexity, point to the need for more-targeted and more-enforceable policy interventions across policy, partnerships, technical capacity, stockpiling, and marketplace incentives.

The analysis is intended for policymakers, industry leaders, and researchers seeking to support a more robust and adaptable domestic PPE manufacturing base for future biological emergencies. It is of particular relevance to those with the authority to shape procurement, stockpiling, and regulatory frameworks, the levers most likely to determine whether domestic SME manufacturers can remain viable outside crisis periods. The recommendations span supply-side financing, demand-side commitments, and regulatory reform and, thus, require coordinated action across federal procurement, emergency preparedness planning, health care purchasing, and small business policy. Finally, we hope this analysis honors the insights and hard-won

experience of those who worked to ensure access to PPE during the COVID-19 pandemic—navigating broken supply chains, regulatory hurdles, and an uncertain market—and who remain committed to strengthening our preparedness for future biological threats. It is important to note that PPE supply chain and access issues have occurred during biological events, as shown in Appendix D, which highlights the need for more resilient PPE supply chains.

Report Overview, Key Findings, and Recommendations

To analyze the landscape of PPE production and sales within the United States in this report, we focused on N95s and EHMRs and the ability of SMEs to surge-manufacture them during, and in anticipation of, large-scale biological crises.

We begin by establishing a foundation for understanding U.S. respirator production and supply, the respirator consumer base, and the relevant policy landscape (Chapter 2). The supply-side discussion outlines key features of N95s and EHMRs and identifies the primary actors within the respirator supply chain. We then characterize the U.S. consumer base, spanning federal agencies, private health care systems, non-health care industries, and individual purchasers. An overview of the policy landscape affecting PPE production and sales provides baseline context for how policies can advance—or possibly hinder—SMEs’ ability to surge-manufacture respirators. Together, this baseline is intended to equip policymakers with the context needed to characterize the challenges affecting SME surge-manufacturing capacity and to identify potential pathways for addressing them.

We then detail the core challenges affecting U.S. SMEs’ ability to scale up respirator production (Chapter 3):

- **Supply-side challenges** include strained access to materials, limited technical capacity, prohibitive up-front and operational costs, and difficulties in recruiting and retaining personnel.
- **Demand-side challenges** include intense competition with large domestic and foreign firms, constrained distribution and sales channels, and volatile demand patterns often compounded by uneven stockpiling practices.
- **Regulatory challenges** include burdensome administrative and compliance requirements and loopholes or noncompliance to regulations that can distort competition.

The chapter also examines adjacent issues affecting PPE use in the United States that emerged during the research, including design limitations that affect PPE usability across demographics, the reluctance of SMEs to invest in innovation, and potential politicization of efforts to promote PPE use.

Finally, we present recommendations to strengthen SME resilience and competitiveness (Chapter 4):

- **Supply-side recommendations** aim to lower up-front costs, enable technical readiness, and attract and sustain skilled workforces.
- **Demand-side recommendations** aim to ensure sustained demand, improve price competitiveness, and promote market fairness.
- **Regulatory recommendations** aim to close loopholes to safeguard fair competition and enable regulatory compliance.

To analyze the U.S. PPE production, consumer, and policy landscapes affecting SMEs, we employed a multitiered methodological approach (detailed in Appendix A). The analysis began with a baseline review of the CMU study on SME barriers during COVID-19, which helped identify gaps in timing and scoping and guided subsequent inquiry. We then conducted an extensive review of academic, industry, and government

literature on respiratory PPE manufacturing and supply chain dynamics, supplemented by semistructured interviews with subject-matter experts in manufacturing, policy, and emergency response between December 1, 2025, and April 3, 2026. The interviews were not for attribution, so no names are provided.

Finally, Appendix B describes relevant COVID-19-era policy; Appendix C explains the types of respirators; Appendix D presents our case study; and Appendix E describes baseline PPE demand.

U.S. Respirator Production, Consumer Base, and Policy Landscape

Addressing the challenges that SMEs are likely to face in scaling PPE manufacturing during, and in anticipation of, future biological crises will require a clear understanding of the market structures and regulatory conditions that shape PPE production and procurement. These factors determine both the constraints SMEs encounter and the feasibility of potential solutions. This chapter provides that foundation by outlining the key elements of respirator manufacturing, tracing the supply chain from material acquisition through distribution, characterizing the major segments of PPE demand, and summarizing the policy landscape governing production and procurement.

N95 Respirators and Elastomeric Half-Mask Respirators

In this report, we focus primarily on firms' capacity to surge-manufacture two types of air-purifying respirators (APRs): **N95 respirators** and **EHMRs**. APRs represent the class of masks that medical, public health, and other essential personnel most commonly used during the COVID-19 pandemic due to their ability to filter airborne particles and block droplets from inhalation.

N95 respirators are disposable air-purifying masks designed to filter out at least 95 percent of airborne particles, providing high protection against infectious aerosols, such as those carrying Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2). Constructed from multiple layers of nonwoven polypropylene (typically including spunbond outer layers and a meltblown electrostatically charged filter core), N95s are lightweight and breathable but are not resistant to oil-based particulates.¹ The U.S. Food and Drug Administration (FDA) regulates N95s as medical devices for health care use and requires proper fit-testing to ensure an effective facial seal.² Their proven filtration efficiency and ease of distribution established them as the standard in health care settings throughout the COVID-19 pandemic.

EHMRs, which provide greater or equivalent protection to most N95s, are reusable respirators featuring a facepiece made of silicone or rubber and detachable filters, which can be replaced as needed.³ Unlike single-use N95s, EHMRs can be disinfected and reused multiple times, providing sustained high-level protection while reducing waste and supply strain during pandemic surges. They can protect against a range of hazards,

¹ Armbrust USA, "What Are N95 Masks Made Of?" webpage, undated; Gregory S. Larsen, Yongqiang Cheng, Luke L. Daemen, Tej N. Lamichhane, Dale K. Hensley, Kunlun Hong, Harry M. Meyer III, Steven J. Monaco, Alan M. Levine, Richard J. Lee, et al., "Polymer, Additives, and Processing Effects on N95 Filter Performance," *ACS Applied Polymer Materials*, Vol. 3, No. 2, January 2021; Sabit Adanur and Ajay Jayswal, "Filtration Mechanisms and Manufacturing Methods of Face Masks: An Overview," *Journal of Industrial Textiles*, Vol. 51, No. 3 Supp., June 2022.

² FDA, "N95 Respirators, Surgical Masks, Face Masks, and Barrier Face Coverings," October 21, 2024.

³ National Institute for Occupational Safety and Health (NIOSH), "Elastomeric Respirators," webpage, U.S. Centers for Disease Control and Prevention, February 3, 2025b.

including airborne particles, gases, and vapors, depending on the filters used. EHMRs often include exhalation valves to decrease breathing resistance; however, some designs now include filtered valves to improve source control.⁴ Their durability and reusability have made EHMRs increasingly attractive for health care and public health response, especially when disposable mask supplies are limited.

Respirator Supply Chains

Respirator manufacturing depends on complex supply chains spanning material acquisition and processing, manufacturing, and distribution. The structure of these supply chains, including reliance on globally sourced inputs, specialized machinery, and concentrated procurement networks, can significantly affect SMEs' ability to acquire materials, scale production, compete for buyers, and sustain operations during periods of heightened demand. Understanding the respirator supply chain can help clarify where supply-side constraints emerge and inform interventions to support SMEs' scale-up capacity.

Material Acquisition and Processing

The primary materials used in respirator manufacturing include polypropylene (for N95 masks and EHMR filters) and elastomeric materials, such as natural rubber, synthetic rubber (including thermoplastic elastomer), and silicone for EHMR facepieces. Polypropylene, a thermoplastic derived from refined petroleum, is produced globally, with China as the largest producer and the United States also playing a significant role. It is meltblown for filtration elements and spunbond for structural elements at manufacturing facilities specializing in nonwoven fabrics.⁵ Although some large respirator manufacturers, such as 3M, process polypropylene in house, SMEs often purchase the fabric wholesale; one such processor is NY Embroidery Studio. The limited availability of meltblown materials and associated machinery often creates a key bottleneck for respirator manufacturers, particularly during a demand surge. Elastomeric materials for EHMR facepieces follow a similarly globalized supply structure, with thermoplastic elastomer—the most commonly used option—produced primarily in China, Europe, and the United States.⁶

Manufacturing

Several large firms, including 3M, Honeywell, Owens & Minor, Moldex, and Prestige America, have historically dominated respirator manufacturing within the United States. These firms' collective production capacity formed the basis of initial White House estimates of PPE supply at the onset of the COVID-19 pandemic. Large manufacturers typically enjoy numerous structural advantages in the U.S. PPE market, including established manufacturing infrastructure, regulatory approvals, and distribution relationships, as well as prior organizational experience scaling domestic production and supplying major medical markets.

A broad ecosystem of SMEs also played a meaningful role in manufacturing respirators during the COVID-19 pandemic; estimates indicate 192 firms participated in manufacturing efforts.⁷ Additional SMEs

⁴ Rohan Fernando, Jeffrey Peterson, and Lee Portnoff, "Advancements in Elastomeric Respirator Technology for Use as Source Control," National Institute for Occupational Safety and Health, U.S. Centers for Disease Control and Prevention, December 31, 2020.

⁵ King, 2024.

⁶ US Masterbatch, "Top TPE Material Manufacturers in China," February 26, 2026.

⁷ This estimate derives from research combining web-scraping data from Thomasnet with FDA and NIOSH approval databases, federal contract records, and D&B Hoovers firmographic data (see Kalathil, Morgan, and Fuchs, 2024).

also participated in adjacent manufacturing activities, including seven firms producing nonwoven fabrics for medical-grade masks and five manufacturing latex-free elastics. Many SMEs entered the market during the pandemic as new entrants with no prior PPE experience, firms pivoting from adjacent manufacturing sectors, or both. SMEs often filled demand that larger incumbents could not reach, particularly in nonhospital markets, such as dental offices, construction sites, food processing facilities, and direct-to-consumer retail.

Distribution

In the United States, large medical supply vendors, such as Medline, are the primary PPE distributors. PPE manufacturers contract with these vendors to supply health care systems. Some distributors work directly with or have incorporated manufacturers to produce their own lines of PPE products. Distributors work closely with group purchasing organizations (GPOs) that manage medical supply purchases. GPOs enable volume-based pricing, allowing them to negotiate lower prices per unit than individual hospitals could achieve. GPOs also streamline the procurement process because they can contract with a variety of individual suppliers under one contract. These organizations also share responsibility for performing quality assurance and regulatory compliance checks on PPE, which reduces the risk of counterfeit or substandard PPE. GPO relationships provide these key benefits to hospitals and health systems but often create barriers for smaller PPE suppliers. As a result, SMEs often do not or are unable to use these supply and distribution systems and, instead, often sell directly to hospitals, companies, or individual consumers.

Respirator Consumer Base

Mask suppliers and intermediary entities distribute PPE to a wide range of public and private consumers, each with distinct procurement practices and needs. Purchasing and demand patterns could shape how SMEs access markets, compete for contracts, and respond to fluctuations in demand during biological crises. Establishing a foundational understanding of the SME-produced PPE market can help identify where market barriers arise and where new policy measures could advance SMEs' ability to scale up respirator production.

Key consumers include the following:

- **Federal health care systems** include the Veterans Health Administration, the Indian Health Service, and TRICARE. PPE procurement for U.S. Department of Veterans Affairs (VA) facilities, the largest federal consumers, is channeled primarily through the Medical/Surgical Prime Vendor Program, which coordinates by bulk purchasing through favored vendors.⁸ Overall, between January 2020 and August 2021, federal agencies spent nearly \$1.5 billion across 7,635 contracts for regulated masks, respirators, and other unspecified mask products.⁹
- **Private health care systems**, the largest consumers of PPE overall, largely acquire PPE through GPOs, which oversee bulk purchase agreements in close cooperation with medical supply distributors. Most hospitals do not directly control PPE procurement, although some smaller health care providers do.
- **Non-health care commercial entities** procure PPE, including respirators, for protecting workers from respiratory pathogens and other respiratory hazards. Construction, industrial, and agricultural workers who work with, or in close proximity to, hazardous substances require PPE. Respirators for nonbiologi-

⁸ Office of Procurement, Acquisition and Logistics, U.S. Department of Veterans Affairs, "Medical/Surgical Prime Vendor Program (MSPV)," webpage, last updated April 20, 2026.

⁹ Tinglong Dai, Muhammad H. Zaman, William V. Padula, and Patricia M. Davidson, "Supply Chain Failures amid Covid-19 Signal a New Pillar for Global Health Preparedness," *Journal of Clinical Nursing*, Vol. 30, Nos. 1–2, January 2021.

cal respiratory hazards are manufactured with requirements different from those for medical use and are regulated by the National Personal Protective Technology Laboratory.¹⁰ Relative to other forms of PPE, respirators are likelier to skew toward nonmedical uses: Before the COVID-19 pandemic, about 90 percent of N95 sales went to industrial and construction settings, with the remaining 10 percent serving the medical market.

- **Federal, state, and academic laboratories** represent a small but consistent portion of the PPE consumer market. Lab personnel require a wide variety of respirators to protect from biological and nonbiological hazardous particles.
- **Individual consumers**, who purchase PPE through public marketplace services, represent a variable share of overall demand, often spiking during public health emergencies. Few respirator manufacturers directly target individual consumers.

Policy Landscape Relevant to Respirator Manufacturing

Government policies play a central role in shaping the production, procurement, distribution, sale, and availability of PPE during and between biological crises. Regulations, domestic sourcing requirements, emergency authorities, and federal procurement practices can together affect how easily firms, and SMEs in particular, can acquire necessary inputs, scale production, enter the market, and compete for buyers. Understanding this policy landscape is essential for identifying the specific challenges SMEs face because many of these barriers stem from how the U.S. federal and state governments design and implement policies. Table 2.1 outlines the key policy frameworks governing PPE manufacturing and procurement in the United States, providing the context to help inform changes to the existing policy landscape to advance SMEs' capacity to surge-manufacture PPE in future crises. Additional policy measures that emerged during the COVID-19 pandemic are detailed in Appendix B.

Three key categories of existing policies or laws could affect SME PPE manufacturing capacity:

- **Domestic sourcing and procurement requirements:** Numerous policies and laws establish requirements and incentives for federal agencies and certain health care providers to purchase PPE manufactured in the United States. These measures often mandate purchases of U.S.-made products in procurement contracts, prioritize long-term buying, or offset cost differences between domestic and imported products. Requirements to purchase U.S.-made PPE can help establish stable demand for U.S. manufacturers, including in between biological crises when demand might diminish. Their effectiveness could vary depending on built-in waivers or exceptions.
- **Policies enabling government intervention in private markets:** Some U.S. measures enable the federal government to directly intervene in private markets during emergencies to ensure the United States can access adequate supplies of critical goods, such as PPE during a biological crisis. Such laws as the Defense Production Act (DPA) and the Stafford Act allow the government to prioritize orders, allocate materials, provide financial support, and direct private industry to produce key goods.¹¹ Although these laws are not specific to PPE manufacturing or SMEs, the DPA and the Stafford Act could directly support SME PPE surge manufacturing by establishing clear demand and reducing market entry barriers during emergencies.

¹⁰ FDA, 2024.

¹¹ Public Law 81-774, Chapter 932, Defense Production Act of 1950, September 8, 1950; Public Law 93-288, Robert T. Stafford Disaster Relief and Emergency Assistance Act, May 22, 1974.

- **Policies enabling SMEs:** Numerous U.S. measures aim to address needs of smaller and medium-sized firms and advance their competitiveness with large businesses. Such policies, which often involve loans, fee waivers, tax incentives, and other mechanisms, help SMEs access capital, lower administrative costs, navigate or overcome regulatory burdens, and compete for federal contracts. SMEs that manufacture PPE could leverage these policies to improve their scale-up capacity during biological crises and enter the PPE market.

TABLE 2.1

Example Policies Relevant to U.S. Respirator Manufacturing

Policy or Law	Summary
Domestic sourcing and procurement requirements	
Buy American Act (BAA)	• Requires federal agencies to purchase domestic materials or products for public use, with exceptions if U.S. options are not available or costs are unreasonable^a
Trade Agreements Act (TAA) of 1979	• Allows waiver of the BAA for “eligible products” from “designated countries” for certain federal contracts, with monetary thresholds • Considers goods as TAA-compliant if they are manufactured or substantially transformed within the United States or a designated country^b
Make PPE in America Act	• Requires certain federal PPE contracts to cover at least two years and PPE to be manufactured within the United States, applying to major U.S. agencies (the U.S. Department of Homeland Security [DHS], the U.S. Department of Health and Human Services [DHHS], and the VA) • Encourages long-term contracts • Allows waivers • Closes some BAA loopholes by specifying minimum U.S.-made content^c
U.S. Small Business Administration (SBA) Set-Asides (Federal Acquisition Regulation [FAR] 25.401(a)(1))	• Forbids TAA waivers of BAA requirements if a domestic small business producer exists^d
Berry Amendment	• Requires the U.S. Department of War (DOW) to procure certain products from fully domestic sources, applying to contracts executed through other agencies on behalf of DOW • Allows waivers, depending on availability and price considerations^e
Kissell Amendment	• Requires DHS to prioritize purchasing textile-based products, including PPE, from U.S. sources but may be superseded by trade agreements • Allows exceptions for quality, availability, price, emergencies, small purchases, and limited foreign content^f
Centers for Medicaid and Medicare Services (CMS) Hospital Payment Regulations	• Allows a hospital participating in Medicare to receive payment adjustments to offset the cost of purchasing U.S.-made N95s versus foreign-produced N95s, if the hospital meets tracking and reporting requirements^g
Policies enabling government intervention in private markets	
DPA	• Allows the President to direct private companies to prioritize federal orders, allocate materials, restrict hoarding, make loans or guarantees, buy products, install equipment, or authorize intercompany coordination for national defense, including PPE during emergencies^h
Stafford Act	• Enables the Federal Emergency Management Agency (FEMA) to provide financial and physical assistance (including PPE procurement and distribution) to state, local, tribal, and territorial governments or nonprofits during declared emergencies • Allows FEMA to cover 75 to 100 percent of PPE costsⁱ

Table 2.1—Continued

Policy or Law	Summary
Policies enabling SMEs	
Made in America Manufacturing Finance Act	Doubles SBA loan limits for small manufacturers up to \$10 million^j
7(a) Working Capital Pilot Program	Provides transaction-based financing for PPE makers to cover 100 percent of direct costs for individual orders, helping SMEs fulfill large contracts^k
Made in America Initiative	Enables fee waivers and onshoring portals, under the SBA, to help small PPE manufacturers reduce borrowing costs and source domestic raw materials^l
One Big Beautiful Bill Act	Allows PPE manufacturers to write off 100 percent of the cost of “qualified” new equipment in the first year^m
Small Business Regulatory Enforcement Fairness Act	- Requires Occupational Safety and Health Administration (OSHA) rulemaking to consider small business input on regulations (such as PPE fit requirements) - Mandates publication of compliance guides - Allows SBA ombudsman to advocate for small firms facing unfair or excessive regulatory enforcementⁿ
Hollings Manufacturing Extension Program and Awards Programs	Supports SMEs through state-designated Manufacturing Extension Partnership (MEP) Centers, connecting PPE producers with health care buyers^o

^a J. Kenneth Fasick, “Statement of J. Kenneth Fasick, Director, International Division, Before the House Subcommittee on Legislation and National Security on the Buy American Act,” U.S. General Accounting Office, April 5, 1978; Public Law 72-428, An Act Making Appropriations for the Treasury and Post Office Departments for the Fiscal Year Ending June 30, 1934, and for Other Purposes, March 3, 1933 (Buy American Act).

^b Gormley Group, “List of TAA Designated Countries,” webpage, undated.

^c Office of the Chief Procurement Officer, U.S. Department of Homeland Security, “Homeland Security Acquisition Regulation, Make Personal Protective Equipment in America Act Restrictions on Foreign Acquisition (HSAR Case 2024-003),” *Federal Register*, Vol. 89, No. 190, October 1, 2024; Public Law 117-58, Infrastructure and Jobs Act, Division G—Other Authorizations; Title IX, Build America, Buy America; Part II, Make It in America; Subtitle C, Make PPE in America Act, November 15, 2021 (Make PPE in America Act).

^d John Holtz, “TAA Can Apply to Small Business Set-Aside: COC Clarifies Trade Agreements Act and Buy American Act Applicability,” *SmallGovCon* blog, August 15, 2025; U.S. Code, Title 19, Customs Duties; Chapter 13, Trade Agreements Act of 1979.

^e Office of Textiles and Apparel, International Trade Administration, “The Berry Amendment—Covered Items,” webpage, U.S. Department of Commerce, undated; Warfighting Acquisition University, “Berry Amendment,” undated; Defense Federal Acquisition Regulation Supplement, Part 225, Foreign Acquisition; Section 225.7002-1, Restrictions (Berry Amendment). The Department of War is designated the Department of Defense under Public Law 81-216, National Security Act Amendments of 1949.

^f Michael Alan Havlin and Alexandra G. Neenan, “Domestic Preference Statutes: The Berry Amendment and the Kissell Amendment,” Congressional Research Service, IF13001, version 3, May 19, 2025; U.S. Code, Title 6, Domestic Security; Chapter 1, Homeland Security Organization, Subchapter VIII: Coordination with Non-Federal Entities; Inspector General; United States Secret Service; Coast Guard; General Provisions; Part H, Miscellaneous Provisions; Section 453b, Requirement to Buy Certain Items Related to National Security Interests from American Sources; Exceptions (Kissell Amendment).

^g Centers for Medicare and Medicaid Services, U.S. Department of Health and Human Services, “Medicare Program; Ensuring Safety Through Domestic Security with Made in America Personal Protective Equipment (PPE) and Essential Medicine Procurement by Medicare Participating Hospitals,” *Federal Register*, Vol. 91, No. 19, January 29, 2026.

^h Anshu Siripurapu, “What Is the Defense Production Act?” Council on Foreign Relations, May 22, 2025. For example, the DPA expanded the supply of meltblown fabric during the COVID-19 pandemic. This also shows how investments upstream of PPE manufacturing can unlock flexibility and surge capacity downstream.

ⁱ Tennessee Department of Education, “Stafford Act and COVID-19 Reimbursement Fact Sheet,” updated June 9, 2020; Public Law 93-288, 1974.

^j SBA, “Administrator Loeffler Applauds House Passage of ‘Made in America Manufacturing Finance Act,’” press release, December 1, 2025c; S. 1555, Made in America Manufacturing Finance Act of 2025, 119th Cong., 1st Sess., introduced by Senator Joni Ernst to the U.S. Senate Committee on Small Business and Entrepreneurship, May 1, 2025.

^k SBA, “7(a) Working Capital Pilot Program,” webpage, last updated February 6, 2026a.

^l SBA, “SBA Launches Onshoring Portal to Advance America’s Economic Comeback,” press release, May 20, 2025a.

^m Public Law 119-21, An Act to Provide for Reconciliation Pursuant to Title II of H. Con. Res. 14, July 4, 2025. Section 70302 allows companies to deduct domestic research and development expenditures immediately or amortized over a period—although not less than 60 months—selected by the company. By contrast, foreign research expenditures must be amortized over a 15-year period. See also U.S. Code, Title 26, Internal Revenue Code; Subtitle A, Income Taxes; Chapter 1, Normal Taxes and Surtaxes; Subchapter B, Computation of Taxable Income; Part VI, Itemized Deductions for Individuals and Corporations; Section 174, Amortization of Research and Experimental Expenditures.

ⁿ OSHA, “Small Business Regulatory Enforcement Fairness Act of 1996 (SBREFA),” webpage, U.S. Department of Labor, undated-b; New Orleans District, U.S. Army Corps of Engineers, “Summary of SBREFA,” undated; Centers for Medicare and Medicaid Services, “CMS Small Business Administration Ombudsman,” webpage, last updated November 12, 2025.

^o National Institute of Standards and Technology (NIST), “Manufacturing Extension Partnership (MEP),” webpage, U.S. Department of Commerce, undated.

Challenges Affecting U.S. Small and Medium-Sized Enterprises in Scaling Up Respirator Manufacturing

U.S. SMEs seeking to scale up PPE manufacturing during biological crises face a set of barriers spanning the full operational life cycle, from sourcing inputs and securing financing to selling and distributing products. Although many of these challenges affect PPE manufacturers broadly, SMEs face distinct structural disadvantages that might make these barriers more difficult to navigate before and during a future crisis. In this chapter, we address challenges affecting SMEs' capacity to surge-manufacture PPE in three broad, interrelated areas:

- **Supply, production, and capacity challenges:** These challenges involve barriers to establishing and operating manufacturing, including access to equipment, materials, and other inputs, as well as the technical capacity and expertise required to produce at scale.
- **Demand and market access constraints:** These challenges include barriers to entry, profitability, enterprise growth within the U.S. PPE sector, and the reliability of a customer base, including through government-procured stockpiling.
- **Regulatory hurdles:** These challenges encompass barriers and complexities associated with complying with regulations, which can limit market entry, slow innovation, and increase operating costs.

The chapter concludes with an overview of adjacent issues that, while not directly relevant to SME PPE surge-manufacturing capacity, represent potential challenges to U.S. PPE use that emerged during the research process. Such issues could warrant policy attention to ensure U.S. citizens can access reliable, high-quality PPE during a future biological crisis.

To identify these findings, we analyzed the CMU report for background and context, conducted a comprehensive literature review using a wide range of sources, and conducted semistructured interviews with subject-matter experts familiar with the PPE manufacturing landscape. These approaches enabled us to capture both documented challenges and real-world perspectives from those directly involved in the industry. Full details on the CMU study methodology are provided in Appendix A.

Supply, Production, and Capacity Challenges

The capacity of SMEs to surge-manufacture PPE at scale depends on numerous interrelated supply-side factors, including supply and material acquisition, technical capacity, prohibitive initial and operational costs, and staffing. Across supply chains, manufacturers and, in particular, SMEs encounter difficulty securing key raw materials, such as polypropylene-derived nonwoven fabrics, particularly during periods of surge demand and global supply disruption. Production also requires highly specialized technical expertise and infrastructure, including precise control over complex fabrication processes for N95 and EHMR respirators. High up-front capital requirements for machinery, facilities, certification, and materials, combined with

ongoing operational costs and limited ability to achieve bulk purchasing efficiencies, further constrain SME scale-up potential. Finally, staffing challenges—including surge hiring difficulties, retention constraints, and limited access to specialized engineers and technical personnel—reduce SMEs’ ability to sustain and expand production capacity under crisis conditions.

Supply and Material Acquisition

U.S. PPE SMEs often face hurdles in securing essential materials due to supply chain complexity and constraints:

- **Key material acquisition:** In periods of high demand, such as biological events, manufacturers often face challenges in sufficiently sourcing polypropylene-derived nonwoven fabrics at scale, affecting manufacturers of both N95s and EHMRs.¹ One interviewee emphasized the complicated nature of the supply chains for synthetic nonwoven fabrics, which is exacerbated by the fact that large portions of these supply chains have been outsourced to low-cost foreign producers. Insufficient U.S. supply of fabric was one of the largest reasons for the mask shortage during the COVID-19 pandemic.²
- **Reliance on global supply chains:** The U.S. PPE industry’s deep reliance on global supply chains, particularly for raw materials and specialized components from China, introduces significant risk. International disruptions, foreign export controls, or geopolitical tensions can swiftly compromise domestic manufacturers’ ability to scale up production during a crisis. For example, China was the source for one-third of disposable face masks and two-thirds of nondisposable face masks used in health care.³

Technical Capacity

SMEs encounter significant technical barriers in PPE manufacturing, including the need for highly specialized equipment, advanced fabrication skills, and careful process control. Challenges affecting SME technical capacity include the following:

- **Producing N95 masks requires highly specialized technical skills and infrastructure.** This is especially so for spunbond and meltblown nonwoven fabrication processes. Achieving target filtration performance requires precise control over multiple interdependent parameters, including electrostatic charge application, fiber layering, thermal or mechanical bonding, and overall material morphology. These requirements present considerable barriers to entry for less experienced manufacturers, including many SMEs. The complexity is further exacerbated when incorporating biodegradable or alternative polymer feedstocks, which may introduce variability in processing conditions and end-use performance.
- **Design, materials, production, and intellectual property requirements make EHMR manufacturing more complex.** EHMR manufacturing is more challenging than N95 production because EHMRs have more components and higher technical requirements. N95s and EHMRs require different machinery and manufacturing processes, so scaling up production of one does not enable scaling up of the other. Several experts noted that intellectual property protections and patents on EHMRs can constrain

¹ Emily Feng and Amy Cheng, “COVID-19 Has Caused a Shortage of Face Masks. But They’re Surprisingly Hard to Make,” NPR, March 16, 2020.

² Tom Simonite, “How Decades of Offshoring Led to a Mask Shortage in a Pandemic,” *Wired*, March 29, 2020.

³ Akinluwa A. Demehin, “AHA Responds to Department of Commerce’s RFI on PPE and Medical Equipment,” press release, American Hospital Association, October 17, 2025.

design and development, while suitable default machinery may be less readily available in the marketplace. Because EHMRs are designed for repeated use with disinfection, materials must be able to withstand repeated cleanings and wear without loss of performance. As with N95 respirators, EHMR manufacturing requires careful control of complex and technical processes, especially when manufacturers produce their own filters.

Prohibitive Initial and Operational Costs

SMEs in the PPE industry are held back by steep up-front and ongoing costs for specialized equipment, facilities, and regulatory compliance, as well as an inability to access bulk purchasing efficiencies. The related challenges include the following:

- **Costs of acquiring and maintaining specialized machinery:** The specialized machinery required to produce nonwoven fibers and manufacture PPE, including both manual and automated options, requires significant costs up front that many SMEs could find prohibitive. Domestic SMEs must often compete directly with large, international suppliers and are increasingly relying on fully automated facilities; one SME interviewee noted that fully automating a facility could cost millions of U.S. dollars. Additionally, PPE manufacturers, who face an often unreliable and inconsistent demand cycle, must either maintain unusable, highly specialized machinery in storage or sell off the machinery for significantly less than the initial purchase price.⁴
- **Significant up-front costs:** Aside from machinery, the up-front costs for acquiring controlled facility space, initial material supplies, regulatory certifications, and trained personnel to operate machinery—especially for nonautomated facilities—are significant.
- **Inability to leverage savings from bulk purchases:** SMEs are often unable to take advantage of cost savings that larger firms might receive when making bulk purchases of industry materials because of a lack of the consistent demand and operational scale required to justify and sustain such large-volume purchasing.⁵
- **Financial vulnerability from production scale-up costs:** Because scaling production requires substantial up-front cash that must be available before revenues are fully realized, SMEs could be especially vulnerable when contracts are reduced or canceled, possibly being left carrying high fixed costs and unused capacity, creating significant financial hardships.⁶
- **Instability of available capital:** Fluctuating prices for raw materials also require SMEs to maintain stable levels of working capital, which is challenging for these organizations.

Staffing

SMEs face persistent staffing challenges, struggling to rapidly hire, train, and retain workers during and after demand surges; contending with limited access to specialized expertise; and facing competition from larger employers. Challenges relating to staffing include the following:

- **Challenges in surge-hiring and training staff:** The COVID-19 pandemic underscored the difficulties PPE SMEs have during emergency periods with hiring skilled workers or training staff to a sufficient

⁴ ManMohan S. Sodhi, Christopher S. Tang, and Evan T. Willenson, “Research Opportunities in Preparing Supply Chains of Essential Goods for Future Pandemics,” *International Journal of Production Research*, Vol. 61, No. 8, 2023.

⁵ Jessica Tagliafierro, “PPE Supply Chain Challenges and Opportunities,” Camoin Associates, October 30, 2020.

⁶ Sodhi, Tang, and Willenson, 2023.

level of competence. SMEs struggle to compete with the wages and benefits that larger PPE producers offer, severely limiting SMEs' ability to recruit top-tier talent during times of high demand.

- **Challenges in retaining staff:** Once surge demand ends, it is nearly impossible for SMEs to maintain staffing levels. For example, the American Medical Manufacturers Association (previously known as the American Mask Manufacturer's Association) estimates that nearly 6,000 of the approximately 8,000 jobs that U.S. mask manufacturers created have disappeared since the peak of the COVID-19 pandemic.⁷ SMEs' ability to retain staff is drastically constrained by the fact that they must compete with cheaper, foreign PPE, resulting in lower revenues (explored further in the "Demand and Market Access Challenges" section).⁸ This limited revenue makes it very difficult to retain workers through industry-standard promotions and pay increases.
- **Limited pool of specialized personnel and technical experts:** According to one PPE manufacturer we interviewed, SMEs often face challenges in securing skilled engineers and technical experts because much of this expertise is concentrated in larger firms or overseas. With smaller workforces, SMEs must frequently rely on generalists rather than process specialists, making it more difficult to manage complex manual tasks, such as assembling and packaging. This reliance places an additional training burden on limited staff and further constrains their capacity to efficiently handle specialized operations.

Demand and Market Access Challenges

The ability of SMEs to participate and thrive in the PPE marketplace is shaped by several interrelated demand-side barriers, including competition, distribution and sales, and reliable demand and stockpiling. SMEs face intense competition from large, established domestic firms and low-cost international producers, which dominate U.S. markets and drive down prices, making it difficult for smaller manufacturers to secure profitable contracts. The highly concentrated distribution landscape, characterized by the dominance of GPOs and major distributors, further limits SMEs' market access; smaller firms often struggle to navigate complex and opaque contracting processes and can face lengthy delays in securing agreements with hospitals and public health systems. Finally, the absence of reliable, predictable demand and the prevalence of boom-and-bust market cycles undermine SMEs' ability to plan for sustained operations and participate effectively in federal and nonfederal stockpiling initiatives.

Competition

U.S. PPE SMEs face steep competition from numerous players, including foreign firms, large U.S. firms that contract with GPOs, non-health care PPE manufacturers, producers of substandard or counterfeit products, and distributors that produce private-label PPE:

- **Competition with foreign firms:** U.S. SMEs face structural cost and scale disadvantages when competing with large international PPE producers, largely from China, Malaysia, Taiwan, Vietnam, and Mexico.⁹ Interviewees noted that these firms, particularly those from China and India, benefit from larger production volumes and broader global customer bases, enabling lower unit costs and more-

⁷ Kirsten Gillibrand, "Gillibrand, Colleagues Urge President Biden, Federal Officials to Purchase and Distribute Made-in-America Masks," press release, January 21, 2022.

⁸ Gillibrand, 2022.

⁹ Tagliafierro, 2020.

consistent demand. As a result, U.S. SMEs, serving smaller and less stable demand pools, struggle to match the lower prices of foreign-produced PPE. The COVID-19 pandemic underscored this dynamic: China accounted for one-half of the global supply of masks, with a daily production of roughly 20 million units prior to the COVID-19 outbreak in 2019, while such nations as the United States and the United Kingdom reported dangerously low levels of PPE supply during the pandemic.¹⁰ During the pandemic, China dominated PPE manufacturing as well, accounting for more than 85 percent of U.S. imports in the category including such products as N95 respirators and both disposable and nondisposable face masks.¹¹ Cheaper imports from China have also led 102 of the 107 PPE companies that were created during the pandemic to stop operations.¹² China still remains the largest face mask manufacturing country after the pandemic with over 36 billion units, and China, South Korea, and India together account for more than one-half the masks produced in 2024.¹³ Constricted supply chains and national export restrictions could severely limit PPE imports from major PPE manufacturing countries, underscoring the need for robust domestic production to reduce foreign dependence on PPE.

- **Competition with large U.S. firms that contract with GPOs:** The U.S. PPE market is highly concentrated among a small number of large firms (3M, Honeywell, and MSA Safety).¹⁴ GPOs, which handle the majority of U.S. hospital procurement, reinforce this concentration by favoring bundled or sole-source contracts with these large vendors.¹⁵ Roughly 70 percent of hospital purchases are made through GPO contracts.¹⁶ SMEs face significant challenges competing with large vendors for these contracts, including lengthy onboarding processes,¹⁷ price disadvantages relative to imports,¹⁸ and limited incentives within distributor- and GPO-driven systems to prioritize SME products over large firms' often discounted products.¹⁹ Furthermore, contracting with GPOs requires SMEs to absorb significant administrative costs that larger firms are likely better positioned to manage. Hospitals are encouraged to increase their purchases through GPOs because of discounts they receive and could be hesitant to buy goods from domestic manufacturers because doing so could reduce their benefits from GPOs.²⁰
- **Competition with non-health care PPE producers:** Despite the surge in PPE demand during the pandemic, the health care sector remained a secondary market for domestic PPE producers, contributing

¹⁰ Talha Burki, "Global Shortage of Personal Protective Equipment," *The Lancet*, Vol. 20, No. 7, July 2020.

¹¹ Ken Roberts, "China More Dominant Than Ever in Covid-Related 'PPE'—and U.S. Flags," *Forbes*, September 19, 2020.

¹² Andrew Jacobs, "Tariffs on China Aren't Likely to Rescue U.S. Medical Gear Industry," *New York Times*, April 24, 2025.

¹³ Market Reports World, *Face Mask Market Size, Share, Growth, and Industry Analysis, by Type (Disposable Masks, Reusable Masks), by Application (Individual, Hospital and Clinic, Industrial), Regional Insights and Forecast to 2033*, Report No. 14719749, April 20, 2026.

¹⁴ Tagliafierro, 2020.

¹⁵ Approximately 96 percent of acute-care hospitals and 98 percent of community hospitals in the United States are affiliated with at least one GPO, and about 70 percent of hospital purchases are made through GPO contracts (Willow Liu Yang, David D. Dobrzykowski, Haitao Li, and William A. Ellegood, "Reducing Hospital Procurement Costs Through Vendor Contract Optimization: A Decision-Support Framework for GPOs," *Production and Operations Management*, Vol. 35, No. 5, May 2026).

¹⁶ Yang et al., 2026.

¹⁷ Interviewees noted that it can take more than one year for an SME to receive a GPO contract, even when they abide by all requirements.

¹⁸ Pradip K. Shukla, Izzet Kosar, and Hedieh Setayesh, "A Literature Review of the United States of America's Reliance on Imported Covid Personal Protective Equipment During the Pandemic," *Journal of Business Research and Reports*, Vol. 2, No. 4, 2025.

¹⁹ Shukla, Kosar, and Setayesh, 2025.

²⁰ Shukla, Kosar, and Setayesh, 2025.

only 17.5 percent of their total revenue.²¹ A higher portion of PPE demand comes from the construction, chemical, and manufacturing industries, where larger producers benefit from scale and lower prices, allowing them to outcompete smaller companies focused on niche medical PPE.

- **Competition with counterfeit or substandard imports and effect on brand reputation:** Counterfeit and substandard imports that fail to meet specifications can erode both the market share and brand credibility of reliable, quality-focused SMEs. U.S. Customs and Border Protection reported a 23-percent rise in counterfeit PPE seizures in 2024, underscoring the scale at which legitimate brands are being undermined.²² In addition, NIOSH investigated 4,163 potentially fraudulent respirator online listings misusing NIOSH certification marks, identifying 1,245 fraudulent listings from May 2025 to January 2026.²³ These products can give consumers a false sense of safety and further detract users from using less-known brands from newer or small domestic SMEs. Moreover, weak vetting processes among retailers and smaller contractors further sustain gray market flow and divert sales away from smaller, more legitimate SMEs.²⁴
- **Competition with distributor-produced private-label and disposable PPE:** SMEs face direct competition from distributors that manufacture and promote private-label PPE, often prioritizing their own products—including disposable PPE—within established hospital and procurement channels and limiting market access for smaller manufacturers.²⁵ One SME manufacturer reported producing respirators that could last up to a year without filter replacement and remain usable in storage for up to ten years. However, they indicated that distributors have little incentive to promote these products because distributors profit more from selling disposable masks that require frequent repurchasing. This dynamic creates a mismatch between distributor incentives and the interests of hospitals and other mask buyers, who may benefit more from longer-lasting options.

Distribution and Sales

SMEs face disadvantages in distributing and selling to U.S. health care markets; entrenched distributor relationships and opaque procurement channels often favor large, established suppliers. Smaller businesses are also particularly vulnerable to logistical disruptions during pandemics and to advertising restrictions, further limiting their market visibility and access to buyers. Challenges relating to distribution and sales include the following:

- **Distributor-controlled access to hospital markets:** SMEs often face structural constraints from distributor-dominated procurement channels. Because most U.S. hospitals contract with specific distributors, SMEs struggle to reach buyers if distributors do not carry their products. In addition, distributors typically require large, consistent order volumes, which SMEs often struggle to meet, given financial and supply chain constraints.
- **Opaque and entrenched distribution channels:** Even when many SMEs overcame production challenges and could sell products at a competitive price, they often struggled to access established dis-

²¹ Targeted News Service, “Rep. Suozzi Travels to DC to Ensure Coronavirus ‘Phase Three’ Passage,” ProQuest, March 31, 2020.

²² Mordor Intelligence, “US Personal Protective Equipment (PPE) Market Size & Share Analysis—Growth Trends and Forecast (2026–2031),” webpage, 2026b.

²³ NIOSH, *eNews: January 2026*, Vol. 23, No. 3, January 5, 2026.

²⁴ Mordor Intelligence, 2026b.

²⁵ Shukla, Kosar, and Setayesh, 2025.

tribution channels. These channels often lack transparency and favor existing relationships, creating information asymmetries that disadvantage less-established firms. As a result, SMEs are often unable to fully leverage a strength: their inherent flexibility to pivot and enter new markets in response to shifting demand.

- **Distribution burdens from pandemic-related restrictions:** During COVID-19, SMEs faced distribution challenges due to nonpharmaceutical interventions, including travel restrictions, limited transport capacity, and reduced workforce availability. These disruptions also contributed to secondary market effects, including material hoarding and PPE profiteering, further straining supply chains. Although SMEs could overcome certain challenges in isolation, the combined impact disproportionately affected PPE manufacturers, necessitating logistical robustness.
- **Reduced market visibility due to advertising restrictions:** Social media and e-commerce platforms, such as Facebook, Google, Instagram, and Amazon, banned PPE advertisements during the COVID-19 pandemic.²⁶ These measures were intended in part to limit purchases by non-health care entities and help ensure that supplies were redirected to health care providers, whose need was most critical. However, the removal of these advertising channels disproportionately affected SMEs that lacked established brand recognition and relied on digital marketing for visibility, limiting their ability to reach potential buyers. Furthermore, the prominence of counterfeit products lessened demand for SME-produced PPE during the pandemic.

Reliable Demand and Stockpiling

Unstable and inconsistent demand, driven by cyclical public health funding and uneven stockpiling initiatives, makes it difficult for U.S. PPE SMEs to sustain production capacity and reliably plan for future market needs. The related challenges include the following:

- **Impact of boom-and-bust cycles on PPE demand:** Public health and health systems funding often follows a boom-and-bust cycle that directly affects PPE demand.²⁷ During public health emergencies, surging PPE demand triggers increased manufacturing and dedicated funding for procurement, as occurred during the COVID-19 pandemic, when emergency appropriations augmented depleted stockpiles and expanded domestic PPE production.²⁸ However, when crises subside, both demand and dedicated funding decline, with a corresponding downturn in PPE manufacturing. Experts we interviewed identified this boom-and-bust cycle as challenging for SMEs because funding instability weakens incentives for companies to maintain capacity between emergencies. This could also undermine SME participation in stockpiling and warm-basing contracts during steady-state periods.
- **Inconsistent demand from federal and nonfederal stockpiling efforts:** The boom-and-bust cycle just noted underscores the funding challenges for public health preparedness. This affects funding for PPE stockpiles as well. Although the federal government has continually funded federal stockpiles, such as the Strategic National Stockpile (SNS), the amount of funding wanes in nonemergency times. Although PPE expires, meaning new PPE product should be procured, the level of dedicated funding in non-emergency times means there are fewer manufacturing partnership opportunities for SMEs with the

²⁶ Targeted News Service, 2020.

²⁷ Matt McKillop, Dara Lieberman, and Rhea K. Farberman, *Public Health Infrastructure in Crisis*, Trust for America's Health, September 2025.

²⁸ Senate Appropriations Committee, "\$340 Billion Surge in Emergency Funding to Combat Coronavirus Outbreak," undated; Mary Denigan-Macauley, *Public Health Preparedness: HHS Should Address Strategic National Stockpile Requirements and Inventory Risks*, U.S. Government Accountability Office, GAO-23-106210, October 2022.

federal government. The federal government has indicated it may work with nonfederal jurisdictions, such as state, regional, and philanthropic entities, to build up PPE stockpiles, which could give SMEs a larger, more reliable customer base. However, experts warn that limited resources will likely hinder these efforts.²⁹

- **Lack of standardized PPE stockpile metrics:** The absence of clear metrics for tracking and monitoring PPE stockpiles limits policymakers' ability to assess preparedness, identify gaps, and design effective, actionable policies, according to one PPE manufacturer. Without consistent measurement, the federal government faces challenges in establishing preparedness targets and supporting accountability. Limited visibility into stockpile levels and potential shortages also constrains manufacturers' ability to anticipate demand and plan production; this includes SMEs, which are especially vulnerable to demand uncertainty.

Regulatory Challenges

Key and interrelated regulatory barriers affect SME participation in the U.S. PPE sector: regulatory and administrative requirements and loopholes or nonadherence to regulations. High certification, testing, and compliance costs, such as those stemming from the NIOSH Respirator Approval Program, require extensive documentation, specialized staff, and ongoing oversight, posing steep hurdles for SMEs with limited resources and capacity. Uncertainty in the regulatory environment, including disruptions to approval pipelines and processes overwhelmed by the influx of counterfeit products, further compounds these challenges and can delay market entry or hinder business for legitimate manufacturers. At the same time, loopholes in trade agreements, waivers favoring low-cost imports, ambiguous product labeling, and inconsistent enforcement of domestic sourcing mandates collectively erode the competitive advantages of U.S.-based SMEs, allowing foreign products to enter the market and diminishing incentives for domestic investment.

Regulatory and Administrative Requirements

The complex regulatory landscape for PPE manufacturing and sales in the United States poses significant challenges to U.S. PPE SMEs, including increasing the administrative burden to meet requirements and diminishing the SMEs' capacity to compete in the PPE market. Related challenges include the following:

- **High certification, testing, and compliance costs:** SMEs must navigate NIOSH's Respirator Approval Program, administered by the National Personal Protective Technology Laboratory,³⁰ which requires that new respirators meet minimum performance and respiratory protection standards prior to market entry. Compliance involves fees (including \$100 per certificate of approval issues), extensive documentation (such as for standard operating procedures [SOPs] and good documentation practices), and ongoing oversight (such as postmarket product evaluations and audits).³¹ Obtaining NIOSH certifications for PPE masks can further involve high up-front costs; long lead times; and limited, expensive testing infrastructure, which can be especially burdensome for SMEs, which often lack dedicated regula-

²⁹ Administration for Strategic Preparedness and Response (ASPR), "Jurisdictional Stockpile Program," webpage, U.S. Department of Health and Human Services, undated-c.

³⁰ NIOSH, "Respirator Certification Fee Schedules and STPs," webpage, U.S. Centers for Disease Control and Prevention, March 4, 2025d.

³¹ NIOSH, "About National Personal Protective Technology Laboratory," webpage, U.S. Centers for Disease Control and Prevention, January 17, 2025a.

tory affairs departments and might have limited up-front capital. Additional requirements, including regulatory filings, fit-testing data, and cybersecurity certification expenses, can further increase costs and complexity, sometimes pushing SMEs to pursue licensing or offload some manufacturing needs to larger companies, limiting SME autonomy.³²

- **Uncertain regulatory environment:** Disruptions to the NIOSH approval pipeline could create significant hurdles for PPE companies seeking accreditation, which highlights how regulatory uncertainty disproportionately affects SMEs, which often have limited bandwidth and resources to respond to unclear regulatory requirements. Even if masks from SMEs do get NIOSH-approved, the cost and administrative burden of fit-testing can make hospitals reluctant to adopt new manufactured products. This creates a paradox: Even when a mask meets regulatory standards, hospitals may still avoid it because fit-testing requires additional resources and coordination. This further reinforces hospitals' choice to keep using the same masks that are being pushed by distributors and GPOs.
- **Prolonged review processes due to proliferation of counterfeit products:** Counterfeit products are widespread, with the U.S. Centers for Disease Control and Prevention (CDC) estimating that about 60 percent of KN95 respirators in the United States are fake. This has strained regulatory oversight and raised quality concerns, including FDA warnings about Chinese-made PPE.³³ The influx of counterfeit products, or even low-quality products, can overwhelm regulatory agencies, slow down certification for legitimate companies, and increase the burden on SMEs to demonstrate product reliability.

Loopholes or Nonadherence to Regulations

Regulatory gaps, exemptions, and inconsistent enforcement of federal rules reduce the effectiveness of domestic sourcing requirements for PPE and undermine the ability of U.S. SMEs to compete fairly in the marketplace. Although the United States has legislation and initiatives that help support domestic sourcing and production of PPE, these programs cannot be effective because of the loopholes in them. Related challenges include the following:

- **Trade agreement provisions allow purchases of foreign PPE:** The TAA allows federal contracts above \$174,000 to waive BAA requirements and allows agencies to treat foreign products as if they were made in the United States. Similar provisions in the Kissell Amendment permit DHS to procure PPE from the more than 120 TAA-designated countries, when consistent with international trade agreements, further reducing demand for U.S.-based SMEs. Importantly, the TAA does not apply to acquisitions set aside for small businesses.³⁴
- **Waivers favoring low-cost imports:** Such provisions as “price unreasonableness” waivers and domestic nonavailability determinations allow agencies to bypass domestic sourcing requirements when foreign products are cheaper or more readily available. Language ambiguities surrounding these mechanisms can further improve the competitive advantage of larger companies that have prearranged agreements at lower prices, squeezing out SMEs.
- **Loopholes in definitions of U.S.-made products:** Such labels as “Assembled in USA” can obscure when companies rely on imported components, making it difficult for SMEs that invest in fully domestic pro-

³² Mordor Intelligence, “Personal Protective Equipment Market Size & Share Analysis—Growth Trends and Forecast (2026–2031),” webpage, 2026a.

³³ Caitlyn Shelton, “CDC Says 60% of KN95 Masks in the US Are Counterfeit: How to Spot Fakes,” Fox 17, January 13, 2022.

³⁴ Code of Federal Regulations, Title 48, Federal Acquisition Regulations System; Chapter 1, Federal Acquisition Regulation; Subchapter D, Socioeconomic Programs; Part 25, Foreign Acquisition, Subpart 25.4, Trade Agreements.

duction to clearly differentiate their products in the market. Despite sounding U.S.-made, products with the “Assembled in USA” label could still contain expensive parts imported from other countries, such as China. For example, the Federal Trade Commission (FTC) brought violations against Axis LED Group LLC and ALG-Health LLC for claiming their products were made fully within the United States despite containing significant amounts of imported materials.³⁵

- **Nonadherence to domestic sourcing requirements in federal channels:** Although some legislation requires federal agencies to buy from U.S. SMEs, inconsistent enforcement has reduced the effectiveness of the legislation in federal purchasing. For example, PPE manufacturers told us that, despite the Make PPE in America Act requiring such federal agencies as the VA to buy domestic PPE, individual VA locations continue to purchase foreign products. As one explanation, GPOs, which largely control purchasing contracts for most U.S. hospitals, are generally not bound by Buy American mandates in their commercial contracts. However, when supplying federally funded facilities, such as the VA, GPOs are subject to these requirements, although waivers remain available. The organizations may, therefore, still negotiate with distributors that source PPE internationally. This may limit the effectiveness of domestic sourcing requirements.

These loopholes collectively undermine U.S. PPE resiliency by weakening the demand signal needed to sustain domestic manufacturing, particularly among SMEs critical to a diversified industrial base. Elements of the TAA and Kissell Amendment neutralize Buy American requirements on higher-value contracts by treating goods from over 120 designated countries as domestic, while price unreasonableness waivers and domestic nonavailability determination provisions enable agencies to default to cheaper foreign sources, favoring large incumbents with prenegotiated import agreements over SMEs. Ambiguous labels, such as “Assembled in USA,” further erode market differentiation for fully domestic producers. Compounding these issues, the Make PPE in America Act is inconsistently enforced, and GPOs play an outsized role in federal procurement despite being largely exempt from Buy American mandates and retaining waiver flexibility even when serving federally funded facilities, such as the VA. As a result, foreign-sourced PPE continues to flow through federal channels. Together, these gaps suppress investment in domestic capacity and leave the United States strategically vulnerable to the same supply shocks exposed during COVID-19.

Adjacent Issues Warranting Attention

During our research and interviews, experts shared observations and findings that fall outside the scope of this project yet provide valuable context for the U.S. PPE landscape and its response to the COVID-19 pandemic. These findings are included in this section for completeness and to identify potential avenues for future research and highlight the wide array of challenges the PPE supply chain faces as a vital resource in pandemic prevention and response, biodeterrence, and preparedness for catastrophic biological events.

PPE Limitations for Particular Demographics

PPE manufacturers have also indicated that there is clear protocol to make all PPE to fit men based on contract restrictions, and little has been done ensure PPE particularly for women and children. Although this is primarily a procurement and standards design issue rather than an SME-specific challenge, it represents a market opportunity that a more diverse and responsive SME manufacturing base could be better positioned to address.

³⁵ McGuireWoods, “FTC Brings Fourth Enforcement Action Under Made in USA Labeling Rule,” August 16, 2022.

Innovation

Multiple experts highlighted a growing need for innovation in respiratory PPE, particularly to address persistent challenges in fit-testing, durability, comfort, ease of use, and the ability to communicate effectively while masked. These challenges affect the industry broadly, spanning hospital systems, manufacturers, and end users alike. In hospital settings, fit-testing is required annually and whenever a new manufacturer or mask model is introduced—a process that can be both costly and time-consuming. When masks are uncomfortable, are poorly fitted, or hinder communication, compliance across all health care settings is likely to suffer. These are not challenges unique to any one segment of the industry but rather systemic issues that underscore the need for better-designed, more user-friendly respiratory PPE. Poor compliance and inadequate fit-testing represent significant public safety vulnerabilities because both directly undermine the protective efficacy of the equipment, regardless of who manufactures it. Although poor compliance and fit-testing failures are primarily public health concerns rather than SME-specific business challenges, they point to an innovation gap that more-specialized manufacturers may be well positioned to fill. This is illustrated by a striking contrast: Individuals not wearing properly fit-tested N95 respirators can transmit an infectious dose of COVID-19 within 25 hours; in comparison, a tightly sealed N95 mask offers 2,500 hours of protection.³⁶ SMEs, less constrained by the standardized production runs that define large-scale manufacturers, may be better positioned to pursue the kind of iterative, user-centered design improvements in fit, materials, and communication-compatible form factors that this innovation gap requires. Addressing these gaps through innovation in materials, design, and fit technology, could meaningfully improve both compliance and protection across the health care system.

Politicization

Although politicization of pandemic protective measures may appear peripheral to manufacturing economics, it functions here as a demand-side variable: When political pressure shapes or overrides public health guidance, it introduces volatility into the consumer and institutional purchasing signals that SMEs, lacking the contractual buffers of larger incumbents, are least equipped to absorb. Multiple manufacturers reported that the politicization of the early COVID-19 response, specifically masking, because of mandates negatively affected and potentially informed recommendations. By early 2022, political support for mask mandates and COVID-19 restrictions had begun to wane even among their traditional proponents. One expert shared a Democratic polling memo submitted to the U.S. House of Representatives that warned that continuing to prioritize COVID-19 precautions and restrictions risked electoral consequences, advising the party to pivot away from pandemic messaging and toward normalcy, given that most Americans had grown fatigued with prolonged mitigation measures.³⁷ Although politicization of pandemic protective measures may seem removed from manufacturing business challenges, demand erosion driven by politically influenced guidance has direct market consequences for SMEs, which—lacking the contractual buffers of large incumbents—are most exposed when public demand signals become unreliable or contested.

³⁶ U.S. District Court, District of Idaho, “Why Cloth Masks Might Not Be Enough as Omicron Spreads,” undated.

³⁷ Molly Murphy and Brian Stryker, “Taking the Win over COVID-19,” memorandum to all interested parties, IMPACT Research, February 24, 2022.

Recommendations to Strengthen the Resilience and Competitiveness of Small and Medium-Sized Enterprises

Ensuring that SMEs have the ability to surge PPE manufacturing in response to a future biological emergency will demand reforms that address the core challenges outlined in Chapter 3: supply, production, and capacity constraints; limitations in market demand and access; and regulatory hurdles. This chapter is organized around these three areas, with each section presenting recommended actions to help SMEs overcome these obstacles. Stakeholders across the PPE production ecosystem, including federal and state policymakers, relevant agencies, and SMEs themselves, can leverage these strategies to strengthen SME resilience and competitiveness and, in turn, enhance the nation's surge capacity for PPE manufacturing.

Although this chapter organizes recommended actions into supply, demand, and regulatory challenges, many options could advance SME surge-manufacturing capacity across multiple dimensions. For example, reforms to PPE stockpiling, despite involving significant government participation and oversight, are treated here as demand-side interventions because stronger stockpiling in between biological crises can create a more predictable and reliable market for SMEs. Similarly, measures that address regulatory features affecting the U.S. PPE market, such as overly broad waiver criteria or imprecise definitions, are categorized as efforts to reduce regulatory barriers, even when they also improve fairness and competition. It is also worth noting that financing recommendations cannot be considered in isolation: SME willingness to take on debt for PPE manufacturing is itself demand-contingent, and without credible demand signals—whether through procurement commitments or stockpiling guarantees—supply-side financing interventions alone are unlikely to motivate SME market entry.

It is important to note that we have not included a cost analysis with these recommendations. They are intended to address the systemic issues identified within the core challenge categories. Hence, any decision to implement them should be paired with detailed cost analysis and implementation planning. Although the initial focus of our research centered on strengthening PPE surge capacity through SMEs, the analysis revealed that supporting SMEs is functionally equivalent to supporting domestic manufacturing. Unlike large multinational corporations, which may offshore production or rely on foreign supply chains, SMEs in the PPE space are predominantly domestic producers by nature. Although offshoring may be less capitally intensive up front for businesses, they may incur higher operational risks, quality control costs, and demand volatility. As a result, policies designed to boost demand for SME-produced PPE serve a dual purpose: They strengthen small businesses while simultaneously advancing domestic manufacturing and supply chain resilience goals. Broader SME participation in the PPE supply chain further reinforces this resilience by creating more-reliable and accessible supplies during periods of high demand or disruption. Supporting SMEs, in this context, is not a separate objective from onshoring production but one of the most direct pathways to achieving it. In short, the recommendations that follow are scoped around improving U.S. PPE manufactur-

ing surge capacity through SMEs and domestic production. Table 4.1 presents an overview of the challenges and associated recommendations that we discuss in this chapter.

Overcoming Supply, Production, and Capacity Challenges

The COVID-19 pandemic underscored how SMEs, even when motivated to address urgent national PPE needs, confronted significant structural barriers to scaling production. Compared with large and established firms in the United States and abroad, U.S. SMEs struggled to secure critical materials and specialized machinery, shoulder substantial up-front capital investments, access procurement advantages that drive down input costs, and attract and retain a skilled workforce. These constraints often left SMEs at a disadvantage, limiting their ability to respond to surges in demand. Although SBA loans can provide valuable opportunities for SMEs, it is important to recognize that taking on debt can be risky for small businesses when future demand is uncertain. Table 4.2 outlines practical strategies for overcoming these supply, production, and capacity challenges, with responsibility for implementation largely resting with Congress and key federal agencies, such as the SBA, ASPR, NIST, and the VA.

Lowering Up-Front Costs

The initial investment required to enter the PPE industry is substantial even for enterprises with existing manufacturing capabilities. Machinery, materials, trained personnel, and certification costs are difficult to manage for SMEs without subsidization. Up-front costs for establishing the automated processes that are critical for SMEs that must compete on price against large, established domestic and international companies are higher still.

Ensure SME Equipment Financing for Automation and Scaling Up

SBA provides desirable financing options to eligible entities, including 504 loans, 7(a) loans, and microloans, along with, in some cases, business counseling and educational support.¹ One recently announced mechanism is the SBA's International Trade Loan Program, which provides loans to small, domestic manufacturers with a 90-percent federal guarantee to "help manufacturers expand facilities, hire workers, and increase

TABLE 4.1
Overview of Challenge Areas and Recommendation Objectives

Challenge Area	Objective of Recommendations
Supply, production, and capacity challenges	• Lowering up-front costs • Enabling technical readiness • Attracting and sustaining skilled workforces
Demand and market access constraints	• Ensuring sustained demand • Improving price competitiveness • Promoting market fairness
Regulatory hurdles	• Closing loopholes to safeguard fair competition • Enabling regulatory compliance

¹ SBA, undated. The term *7(a) loans* refers to a program that provides long-term financing from a variety of sources. These loans are delivered by SBA 7(a) lenders. The term *504 loans* refers to a program that provides long-term, fixed-rate financing via available community-based SBA certified development companies. The term *microloans* refers to a program that provides loans of \$50,000 or less to help businesses and certain nonprofit childcare centers. These loans are provided by intermediary lenders.

TABLE 4.2

Options to Address SME Supply, Production, and Capacity Challenges

Recommendation	Description	Responsible Stakeholders
Lowering up-front costs		
Ensure SME equipment financing for automation and scaling up	Provide SMEs with affordable financing options, such as 504, 7(a), and microloans, to upgrade equipment and increase production capacity ^a	SBA
Enabling technical readiness		
Support SME warm-basing and prepandemic agreements	Maintain continuous production readiness and workforce training during nonemergency periods through advance purchase commitments and reserve training programs	ASPR ^b SNS Congress ^c
Expand technical assistance programs	Offer specialized support to SMEs to assess automation readiness, find suitable high-speed technologies, and build road maps to increase competitiveness against imports	NIST MEP Centers
Attracting and sustaining skilled workforces		
Incentivize employee retention via tax credits	Offer tax credits to SMEs to retain skilled workforce beyond certain tenure benchmarks	SBA
Offset costs for recruiting critical manufacturing personnel	Provide wage subsidies or supplements to SMEs to help recruit essential technical and quality-focused personnel, including skilled technicians, quality engineers, and regulatory specialists	SBA
Tie employee retention grants to PPE production contracts	Offer grants to SMEs for maintaining a consistent workforce while fulfilling federal PPE contracts, supporting long-term personnel stability	VA DHHS

^a SBA, “Loans,” webpage, undated.

^b ASPR, “Industrial Base Management and Supply Chain Office of Critical Medical Equipment,” U.S. Department of Health and Human Services, undated-b.

^c Committee on Appropriations, U.S. House of Representatives, *Departments of Labor, Health and Human Services, and Education, and Related Agencies Appropriations Bill, 2026*, Report 119-271, September 11, 2025.

production.”² The SBA should prioritize making these options easily accessible to PPE SMEs. Such support enables them to upgrade equipment for further automation and process efficiencies, increasing their overall production capacity to meet surged demand during biological crises.

Enabling Technical Readiness

Maintaining a sufficient level of technical capacity in PPE manufacturing during periods of relatively low demand is critical to emergency response. The resources required for manufacturing processes, such as specialized machinery, material stocks, and technical expertise, are reasonably accessible only during non-emergency periods—during emergencies, these resources become tight bottlenecks. Additionally, if left unsupported, these capacities degrade over time as companies shed redundancies and unused capacity. The COVID-19 pandemic demonstrated that, while PPE suppliers will eventually be able to ramp up production to meet surge demand, the process takes too long if the starting point is close to baseline. Technical readiness requires intentional public funding and operational support.

² SBA, “SBA Announces New ‘Made in America Loan Guarantee’ to Restore Manufacturing Dominance,” press release, March 31, 2026b.

Support SME Warm-Basing and Prepandemic Agreements

One method of supporting technical readiness in PPE manufacturing is the establishment of warm-basing, the government support of standby manufacturing capacity. Warm-basing initiatives can take the form of direct subsidies to manufacturers to acquire and set up additional production lines, financial support to maintain a large pool of trained staff, or incentives to maintain material and equipment stockpiles.³ Consistent federal funding should be provided for this purpose to SMEs that demonstrated the ability to surge manufacturing during the COVID-19 pandemic. General industry support via warm-basing initiatives is recommended, but SME technical capacity, being particularly prone to degradation, should specifically be supported. Congress has previously directed funding toward respirator manufacturing warm-basing through ASPR.⁴

Expand Technical Assistance Programs

Technical assistance programs, such as NIST's MEP program, are crucial for new SME manufacturers.⁵ Such support is especially pivotal for PPE SMEs because of the high levels of specialization and automation within the industry. During the COVID-19 pandemic, MEP provided substantial support to domestic PPE manufacturers, including consulting services, product development, return-to-work guides, and assistance on pivoting production processes.⁶ MEP and similar programs should be expanded to support PPE SMEs even during nonemergency periods to help maintain a healthy manufacturing ecosystem and essential manufacturing expertise.

Attracting and Sustaining Skilled Workforces

SMEs often find it challenging to attract and sustain a skilled workforce. For PPE SMEs, which must navigate precipitous drops and towering peaks of demand, this is an immense challenge. When insufficiently supported during nonemergency periods, SMEs lay off huge portions of their staff that must then be rapidly replaced during the next demand spike. The limited workforce is highly concentrated in established companies that are able to provide higher, more consistent pay and benefits, so SMEs must often rely on those with generalist manufacturing expertise—workers who switch to PPE manufacturing positions in response to surge-related job openings.

Incentivize Employee Retention via Tax Credits

Retaining skilled employees is essential for manufacturing specialized products, such as PPE. As the COVID-19 pandemic demonstrated, it is not possible to find and onboard a full complement of skilled workers rapidly in response to emergency conditions. Governments can incentivize retention through financial support to SMEs that maintain operational levels of tenured staff. We recommend using tax credits to accomplish this because they do not require diverting funds for the purpose.

Offset Costs for Recruiting Critical Manufacturing Personnel

Skilled personnel in PPE manufacturing are employed predominantly by established suppliers (e.g., 3M) at salary and benefit levels that SMEs are unable to consistently meet because of their turbulent financial states. Because of this, many SMEs struggle to attract the experienced technical workers who are critical in new PPE

³ Blueprint Biosecurity, *Towards a Theory of Pandemic-Proof PPE*, June 2024.

⁴ Committee on Appropriations, U.S. House of Representatives, 2025.

⁵ NIST, undated.

⁶ NIST, "Gearing Up: Manufacturing Extension Partnership Center Responses to the COVID-19 Pandemic," webpage, updated February 3, 2025.

enterprises. Governments could aid SMEs by offsetting the costs associated with wages, benefits, and promotions that are required to compete for skilled labor with larger suppliers. Favorable loans as provided by the SBA or direct subsidies for this purpose would help SMEs to remain competitive within PPE manufacturing.⁷

Tie Employee Retention Grants to PPE Production Contracts

Contracts to supply public health systems with PPE have been the primary income vehicle for SMEs. Tying these contracts to employee retention measures would push SMEs to maintain a more stable workforce. If such a requirement is enacted, governments must supply some sort of direct financial benefit to compliant SMEs to not overburden them.

Overcoming Demand and Market Access Constraints

As private enterprises, SMEs require clear demand signals, established customer bases, and predictable markets to flourish and remain viable, including in periods between biological crises. Even if SMEs are able to maintain skilled workforces, achieve necessary technical capacity, and generate adequate supply, the sporadic and unpredictable nature of severe biological risks makes effective planning and meeting market demand highly challenging. Additionally, SMEs face significant competition from dominant market players and foreign suppliers capable of producing PPE at lower costs, while the widespread presence of counterfeit products erodes trust in products from lesser-known firms. Table 4.3 offers an overview of policy options designed to address these challenges, including stable procurement mechanisms, such as government stockpiling, initiatives to enhance price competitiveness, and measures that promote fairness in the marketplace.

Ensuring Sustained Demand

Sustained demand is a critical foundation for maintaining a robust domestic PPE supply chain: Manufacturers, particularly SMEs, require reliable purchasing commitments to justify long-term workforce and capital investments. Without predictable demand, domestic producers struggle to remain price competitive against foreign suppliers and are vulnerable to market fluctuations that can force them to scale back or cease production altogether. Policy mechanisms, such as long-term federal contracts, hospital sourcing mandates, and strategic stockpiling reforms, are therefore essential not only to stabilize the market for domestic PPE manufacturers but also to ensure that adequate supplies remain available before the next public health emergency.

The U.S. government has long been a significant purchaser of PPE. It is key that the federal purchasing of PPE continues to maintain the SNS to maintain the stockpiles at DOW and the U.S. Department of Agriculture and to ensure adequate supplies of PPE at federally operated health care entities, such as those within the VA and DHHS systems.⁸ However, there are also opportunities for states and other jurisdictions to invest further in their own stockpiling efforts, including widespread creation of permanent state-operated stockpiles. This would help support a stable PPE customer base for domestic manufacturers, including for small and medium-sized ones. Doing so also has practical benefits. The SNS is designed to provide short-term support to states on request and deploys assets from warehouses in locations that are not publicly disclosed.⁹ Non-

⁷ SBA, 2026b.

⁸ Clifton G. Chappell and Roderick Gainer, *Defense National Stockpile Center: America's Stockpile: An Organizational History*, Defense Logistics Agency, undated; Animal and Plant Health Inspection Service, "National Veterinary Stockpile," webpage, U.S. Department of Agriculture, last updated December 21, 2025; Centers for Medicare and Medicaid Services, U.S. Department of Health and Human Services, 2026.

⁹ ASPR, "Center for the Strategic National Stockpile," webpage, U.S. Department of Health and Human Services, undated-a.

TABLE 4.3
Options to Address Demand and Market Access Constraints

Recommendation	Description	Responsible Stakeholders
Ensuring sustained demand		
Require hospital acquisitions of SME-produced PPE	Mandates that hospitals receiving Medicare and Medicaid funding source a set percentage of PPE from SMEs and maintain a minimum 30-day supply	CMS ASPR
Reevaluate federal stockpiling strategy	Shifts federal stockpiling (SNS and DOW) to such approaches as vendor-managed inventory (VMI) and warm-basing for lower cost, better rotation, and consistency	DHS (ASPR, FEMA) DOW
Advance nonfederal stockpiling	Establishes both state and federal government responsibility for half of needed PPE stockpiles, reducing federal burden and encouraging greater state investment	State health departments Emergency management agencies Governors
Stabilize demand via long-term contracts	Uses long-term federal contracts to stabilize PPE demand and allows SMEs to make durable workforce and capital commitments	DHHS DOW DHS
Improving price competitiveness		
Increase tax incentives for health care to buy from domestic SMEs	Offers federal and state tax incentives, tiered or capped, to health care entities purchasing PPE from SMEs	DHHS Internal Revenue Service SBA CMS State legislatures
Incorporate tariffs on foreign-made PPE	Implements or increases tariffs on imported PPE to enhance the price competitiveness of domestic products	Congress U.S. Trade Representative U.S. Department of Commerce and supporting entities (U.S. Customs and Border Protection, DHS, DHHS, FEMA)
Promoting market fairness		
Ban GPO exclusionary contracting and require transparency	Prohibits GPO use of contract structures or bundling that disadvantages SMEs Require transparency in contracting, pricing, and vendor selection	FTC DHHS Congress U.S. Department of Justice CMS
Insulate VA procurement from vendor influence	Places PPE purchasing authority with VA administrators and clinicians, not supplier representatives or GPO intermediaries	VA Office of Management and Budget Congress

federal jurisdictions possessing their own stockpiles would make more countermeasures readily available to populations in need and create a more distributed network of stockpiled materials. Executive Order (EO) 14239 calls for states and localities to take a more active role in preparedness, signaling the federal government's appetite for states and other jurisdictions to take a more active role in stockpiling countermeasures, such as PPE.¹⁰

Require Hospital Acquisitions of SME-Produced PPE

One key component to ensure sustained demand for domestically produced PPE is to require hospitals that receive CMS reimbursement to source a set percentage of their PPE from SMEs. By tying procurement requirements to federal health care funding, this policy would create a reliable and recurring customer base

¹⁰ EO 14239, "Achieving Efficiency Through State and Local Preparedness," Executive Office of the President, March 18, 2025.

for domestic SME manufacturers, helping them maintain stable operations outside emergency surge periods. Additionally, mandating that these hospitals maintain a minimum 30-day PPE supply would reduce the risk of critical shortages during public health crises and reinforce consistent demand signals that allow smaller manufacturers to plan and invest with greater confidence. Additionally, hospitals resourcing PPE from local or regional SMEs would reinforce national efforts for pandemic prevention and readiness and strengthen these efforts at the state level to alleviate dependence on federal stockpiles.

Reevaluate Federal Stockpiling Strategy

Federal stockpiling strategy could be reevaluated to optimize alternative management models, such as VMI, warm-basing arrangements, or similar approaches. These approaches can support stock rotation, more-consistent procurement cycles, and quick deployment times. The SNS halted VMI contracts in 2017, citing costs and meaningful impacts on preparedness, but reestablished VMI contracts in response to the COVID-19 pandemic.¹¹ The SNS also employs user-managed inventory—prepositioned caches designed for emergency release in less than 12 hours.¹² The SNS currently uses VMI and user-managed inventory approaches in conjunction with SNS-managed inventory.¹³

Public information about which countermeasures in the SNS are currently held under alternative inventory management models is limited. However, a 2023 report from the U.S. Government Accountability Office (GAO) showed that VMI contracts were executed for PPE, ventilators, and other countermeasures in response to the COVID-19 pandemic using pandemic relief funds.¹⁴ Additionally, past research identified highly specialized products that are needed quickly (e.g., antineutropenic cytokines for radiological injuries or CHEMPACKs for chemical injuries) were managed under alternative management models, such as VMI and user-managed inventory.¹⁵

Given that some forms of PPE, such as EHMRs, are more specialized and could be needed quickly in response to a high-consequence public health emergency, such as a respiratory pandemic, they may be candidates for alternative inventory management models. These models could represent additional opportunities for PPE SMEs to contract with the federal government. Therefore, the viability of further adopting these models should be explored.

ASPR, which houses the SNS program, could evaluate whether such alternative inventory management approaches for specialized forms of respiratory PPE are advantageous economically and for national preparedness. Furthermore, ASPR might analyze whether SMEs could serve as viable partners to fill these orders. These assessments could be included in the next Medical Countermeasures Preparedness Review (previously referred to as the SNS Annual Review), a statutorily required annual report that is prepared by ASPR, reviewed by the Public Health Emergency Medical Countermeasures Enterprise, and submitted to Congress.¹⁶

¹¹ Todd Kuiken and Frank Gottron, *The Strategic National Stockpile: Overview and Issues for Congress*, Congressional Research Service, R47400, version 4, updated September 26, 2023.

¹² Kuiken and Gottron, 2023.

¹³ Kuiken and Gottron, 2023.

¹⁴ Denigan-Macauley, 2022.

¹⁵ C. Norman Coleman, Chad Hrdina, Rocco Casagrande, Kenneth D. Cliffer, Monique K. Mansoura, Scott Nystrom, Richard Hatchett, J. Jaime Caro, Ann R. Knebel, Katherine S. Wallace, and Steven A. Adams, “User-Managed Inventory: An Approach to Forward-Deployment of Urgently Needed Medical Countermeasures for Mass-Casualty and Terrorism Incidents,” *Disaster Medicine and Public Health Preparedness*, Vol. 6, No. 4, December 2012; Kuiken and Gottron, 2023.

¹⁶ ASPR, *2024 Public Health Emergency Medical Countermeasures Enterprise (PHEMCE) Strategy and Implementation Plan (SIP)*, U.S. Department of Health and Human Services, September 2024.

Stockpiling production capabilities rather than just goods is also another emerging capability that has gained traction in the defense industrial base context. In this context, warm production lines, maintained tooling, workforce retention, and reserved manufacturing capacity are increasingly viewed as strategic assets rather than just physical inventories of critical materials, intermediate components, or end products. This strategy may also reduce the wastage and obsolescence of expired materials and reduce physical storage space. Stockpiling productive capacity preserves the latent ability to surge output when needed, offering a more flexible and durable form of resilience.

Advance Nonfederal Stockpiling

Expanding nonfederal stockpiling initiatives could create a more robust distributed network of PPE supplies, yield additional stable customers for SMEs, and strengthen alignment with current federal priorities. In response to EO 14239, ASPR's SNS office launched the Jurisdictional Stockpile Program.¹⁷ This program provides technical assistance, stakeholder engagement and liaison services, and exercise support to nonfederal jurisdictions as they work to stand up their own stockpiling initiatives.

To take a larger role in PPE purchasing and stockpiling, states will require monetary, personnel, and operational resources. There are mechanisms states could explore to reduce the financial burden, however. Establishing a one-to-one federal-state matching framework could help balance stockpiling responsibilities across levels of government. Under this model, states would be responsible for procuring and managing half of their required PPE, with federal stockpiles fulfilling the remaining need, creating shared accountability while reducing federal burden and encouraging greater state investment in preparedness. This framework could be managed through the Jurisdictional Stockpile Program and designed to run for a defined number of years, with the goal of providing support to states as they stand up their own stockpiling efforts before assuming full responsibility for their respective stockpiles. As they build and maintain their stockpiles, states should prioritize working with SMEs, especially if SMEs are located within their respective jurisdictions. This will create a steadier manufacturing pipeline for SMEs and reduce reliance on a small number of larger providers. ASPR's Jurisdictional Stockpiling Program could support the inclusion of SMEs in these efforts through stakeholder engagement and liaison services.

States could also explore joining with neighboring states for a regional approach, which may be particularly attractive for smaller states to offset stockpile operational costs. To further mitigate operational costs, state or regional stockpiles could develop programs that extend product shelf life, modeled on the FDA's Shelf-Life Extension Program (SLEP).¹⁸ SLEP evaluates certain federal medical countermeasures to assess whether their expiration dates can be safely extended. Creating SLEP-like programs for state or regional stockpiles would provide a mechanism for reducing costs while still expanding the customer base for PPE manufacturers.

Demand Stabilization via Long-Term Contracts

Federal procurement policy can serve as a demand-side stabilization tool by committing government purchasing power to long-term contracts with domestic manufacturers, especially SMEs. Rather than relying on volatile market conditions, this approach directly addresses the structural barrier of revenue uncertainty that prevents manufacturers from making durable workforce investments. By guaranteeing a sustained demand signal through multiyear procurement agreements, the federal government can incentivize employers to offer stable, long-term employment commitments, translating public purchasing decisions into measurable

¹⁷ EO 14239, 2025; ASPR, undated-c.

¹⁸ FDA, "Expiration Dating Extension," webpage, February 25, 2025.

labor market outcomes. This positions procurement not merely as a transactional function but as a lever for industrial and workforce policy.

Improving Price Competitiveness

Federal and/or state tax incentives could provide a valuable mechanism for supporting domestic PPE SMEs. Such incentives could help drive larger entities, such as hospitals and health care systems, to buy from smaller manufacturers. Incentives could have a certain minimum purchasing percentage threshold or be tiered with a cap, such that a larger percentage of annual purchasing from an eligible SME equates with a larger tax incentive. Even smaller entities, such as private health care practices, could be encouraged to buy from SMEs if tax incentives were designed in an attractive way and benefited operations their size. PPE manufacturers told us that tighter CMS reimbursements on domestically manufactured medical products can also better incentivize hospitals to purchase from domestic PPE manufacturers.

Tariffs and Incentives on PPE Products Made Abroad

Foreign-made PPE, particularly from large-scale manufacturers in countries with lower labor and production costs, has long undercut domestic SMEs on price, making it difficult for local producers to compete and sustain operations. Without intervention, market forces alone are unlikely to shift purchasing behavior in favor of domestic suppliers, leaving the United States vulnerable to supply chain disruptions during public health emergencies. Trade policy and fiscal instruments offer two complementary levers to address this imbalance: Tariffs can raise the relative cost of foreign-made PPE to reduce the price gap, while targeted tax incentives can reduce the effective cost burden on health care entities that choose to source domestically. Together, these mechanisms create both supply-side protection and demand-side incentives that support a more diversified, resilient, and self-sufficient domestic PPE manufacturing base.

Tax Incentives for Health Care to Buy from Domestic SMEs

Federal and state governments should establish targeted tax incentive structures for health care entities that source from SME manufacturers, especially those domestically producing. Incentives would be tiered or capped to ensure fiscal responsibility while prioritizing smaller producers who are most vulnerable to being undercut by large-scale or foreign competitors. This mechanism shifts purchasing behavior within the health care sector by reducing the effective cost of choosing domestic SME suppliers, creating a demand-side subsidy that does not require direct government procurement. The policy acknowledges that SMEs often cannot compete on price alone and uses the tax code as a corrective instrument to level the competitive landscape and sustain a diversified, resilient domestic PPE supply chain.

Tariffs on Foreign-Made PPE

Currently, there are Section 301 tariffs for masks and respirators with surgical and nonsurgical face masks at 50 percent, while there is a 25-percent tariff on disposable textile face masks.¹⁹ The federal government could also potentially put additional tariffs on PPE to help make domestically manufactured PPE more cost competitive with foreign-made PPE masks, according to one manufacturer. The United States eliminated the duty-free *de minimis* exemption for low-value imports from China and Hong Kong for goods that are valued

¹⁹ Crowell, “Answering the Top Seven Questions About Pending Section 301 Deadlines,” webpage, April 7, 2026. Section 301 tariffs are trade remedies imposed under Section 301 of the Trade Act of 1974, which are structured to address “unfair” trade practices that impede or burden U.S. commerce. See also David E. Bond, Matt Solomon, and Ian Saccomanno, “United States Finalizes Section 301 Tariff Increases on Imports from China,” White & Case LLP, September 17, 2024.

below \$800.²⁰ This change can perhaps go further by excluding all masks and respirators from the *de minimis* treatment.

Promoting Market Fairness

Sales contract acquisition and distribution presented immense challenges during the COVID-19 pandemic that have only intensified since because of dwindling demand. Interviewees told us that addressing the root cause of the inconsistent PPE demand signal for SMEs will require prioritizing larger orders and consistent contracts. This will help ensure demand is maintained and the manufacturers can continually make PPE based on these contracts while maintaining the capacity to increase volumes during acute crises.

GPOs and major medical supply distributors dominate private health care PPE procurement and structurally favor large vendors, effectively excluding PPE SMEs from the market and concentrating supply chain risk in a narrow base of large suppliers. In addition, there has been little research on GPOs and the role they play in steering product choice for affiliated hospitals.²¹ This is largely due to data transparency issues: Contracts between hospitals, GPOs, and their associated manufacturers are considered to be trade secrets.²² We were told that, to maintain demand, there also needs to be a push in compliance to hold GPOs accountable to give a threshold percentage of PPE business to domestic SMEs.

Policymakers must pursue targeted regulatory reform to address GPOs and medical distributor contracting practices that create structural barriers limiting fair competition for SMEs in health care PPE procurement.

Ban GPO Exclusionary Contracting and Require Transparency

GPOs should not be allowed to use prohibitive contract structures, exclusivity arrangements, or bundling practices that effectively prevent PPE manufacturers, particularly domestic SMEs, from competing for hospital contracts on merit. The terms for GPO contracts for critical supply categories should be limited and should include more oversight to ensure that new SMEs can be regularly incorporated. If there are qualified alternatives among SME products, then there should be a requirement for multivendor contracts. CMS could condition Medicare and Medicaid reimbursement eligibility on hospitals' demonstrating procurement through multiple vendors. Potentially, GPO contracts could include at least one SME supplier meeting defined qualification standards. Furthermore, bundling critical supply categories with noncritical supply categories should be more heavily regulated. Critical supply procurement needs to be evaluated on its own set of standards and cannot be contingent on other purchasing categories. Transparency in GPO contracting, pricing, and vendor selection criteria should be required. There should be a full breakdown of administrative fees that the GPO collects from vendors.

Insulate VA Procurement from Vendor Influence

The VA health system's PPE procurement decisions should be insulated from vendor influence. Purchasing authority should rest with VA administrators and clinicians and not be shaped by supplier representatives or GPO intermediaries with financial interests in the outcome. VA participation in any GPO PPE contract should meet VA-specific standards. For example, there should be minimum requirements for supplier diversity and domestic content, and pricing structures should be transparent. There should be greater emphasis on

²⁰ White House, "President Donald J. Trump Closes De Minimis Exemptions to Combat China's Role in America's Synthetic Opioid Crisis," fact sheet, April 2, 2025.

²¹ Emma Boswell Dean, Reekarl Pierre, Samuel Carter, and Amelia M. Bond, "Role of Supply Chain Intermediaries in Steering Hospital Product Choice: Group Purchasing Organizations and Biosimilars," *Health Affairs Scholar*, Vol. 2, No. 6, June 2024.

²² Dean et al., 2024.

the VA including a minimum number of qualified suppliers, diminishing the risk of prohibiting sole-source arrangements. The VA should also negotiate agreements that preserve its right to purchase off-contract items from veteran-owned suppliers without forgoing pricing benefits in other categories.

Overcoming Regulatory Hurdles

The U.S. PPE market operates within a vast and complex regulatory landscape, with policies aiming to ensure product quality and protect consumers. However, we found that some aspects of this environment, including broad waiver criteria, vague definitions, and opaque certification processes, can inadvertently disadvantage SMEs. These regulatory deficiencies could favor larger firms and complicate SMEs' ability to access markets and compete for government contracts, ultimately destabilizing demand and narrowing procurement opportunities. Table 4.4 outlines strategies stakeholders can use to help SMEs overcome regulatory barriers, such as closing loopholes and enhancing SMEs' capacity to comply with regulations.

Closing Loopholes to Safeguard Fair Competition

The federal government needs to step up to help GPOs and Medline comply with the Make PPE in America Act. Despite existing statutory requirements that federal agencies procure domestically manufactured PPE, covered agencies—including DOW (Berry Amendment), DHS (Kissell Amendment), and agencies subject to the BAA and the Make PPE in America Act—frequently exploit categories of loopholes to source PPE from offshore suppliers.

Tighten Waiver Criteria to Require Total Cost of Ownership Analysis, Not Only Unit Price

Federal procurement regulations should be updated so that domestic sourcing waiver determinations incorporate total cost of ownership (TCO) analysis for procurements above a certain threshold rather than relying on unit price comparisons. This TCO analysis should account for supply chain risk, quality assurance costs, transportation and logistics, inventory carrying cost, and strategic cost of foreign supply dependence. This would diminish the structural biases that occur in current waiver criteria that systemically disincentivize domestic manufacturers and SMEs. Offshore producers benefit from economies of scale, lower regulatory costs, lower labor costs, and government-subsidized costs but also have hidden costs and potential quality drawbacks and pose strategic vulnerabilities. An existing FAR framework applies evaluation factors (percentage additions) to foreign offers; these factors may still be insufficient proxies for true TCO analysis.²³

Contracting officers should be required to complete TCO analyses for PPE procurements that meet certain threshold criteria, and price unreasonableness waivers should not be allowed unless TCO analysis demonstrates that domestic sourcing is in fact cost prohibitive. There should be a binding TCO methodology that is presented by the FAR Council and other coordinating agencies. The FAR Council should also establish mandatory TCO cost categories and calculation standards that are regularly updated as they apply. A structured risk framework should be developed and put in place that may align with DOW's existing supply chain risk management methodologies.

²³ Code of Federal Regulations, Title 48, Federal Acquisition Regulations System; Chapter 1, Federal Acquisition Regulation; Subchapter D, Socioeconomic Programs; Part 25, Foreign Acquisition, Subpart 25.1, Buy American—Supplies; Section 25.105, Critical Components and Critical Items.

TABLE 4.4
Options to Address Regulatory Hurdles

Recommendation	Description	Responsible Stakeholders
Closing loopholes to safeguard fair competition		
Tighten waiver criteria to require TCO analysis, not only unit price	Reform procurement waiver criteria to require consideration of TCO, ensuring a fairer evaluation of bids from U.S. SMEs	Congress FAR Council
Establish clear, uniform definitions for what constitutes <i>domestically manufactured</i> for PPE	Ensure that only products involving substantial transformation and significant value-added production in the United States qualify as domestically manufactured, helping U.S. SMEs compete fairly for government contracts	Congress Executive Office of the President SBA DHS DHHS VA DOW
Require public reporting of all PPE procurement waivers and opt-outs with detailed justification	Increase the transparency of waiver decisions, enabling SMEs to identify procurement opportunities and challenge exclusions from domestic contracts	Congress FAR Council General Services Administration DHS DHHS VA Made in America Office
Require independent audits of PPE waiver patterns and nonavailability determinations and establish enforceable consequences for agencies that do not meet these standards	Direct GAO and agency inspectors general to review waiver practices and refer suspected fraud, ensuring that SMEs are treated fairly in contracting	GAO Inspectors general Made in America Office
Enabling regulatory compliance		
Form collaborative SME networks for compliance resources	Encourage SMEs to jointly develop and share standardized quality control checklists, SOP templates, and equipment validation protocols to streamline NIOSH compliance and improve market access	SMEs American Mask Manufacturers Association
Develop and publish accessible PPE compliance guidance	Direct NIOSH and the FDA to issue clear certification checklists, equipment validation benchmarks, and SOP frameworks to help SMEs efficiently meet regulatory requirements and compete fairly	NIOSH FDA

Establish Clear, Uniform Definitions of the Term *Domestically Manufactured* for PPE

A uniform federal definition of the term *domestically manufactured* for PPE would establish process-based standards for each PPE category, including respirators, from the earliest feasible material processing state through the finished product. This would replace the current patchwork of inconsistent cost-percentage tests and substantial transformation standards with clear, auditable criteria incorporating the depth of the supply chain and applied uniformly across all federal agencies. This would bridge the different standards set by the Kissell Amendment, Berry Amendment, BAA, and Make PPE in America Act into a uniform standard definition. This definition would apply, with exceptions, to the TAA-designated countries for PPE deemed critical to public health safety and national security. Vendors should be required to submit supply chain disclosures that are subject to audit. An interagency board should review and update category-specific process standards regularly.

Require Public Reporting of All PPE Procurement Waivers and Opt-Outs with Detailed Justification

Federal agencies should report instances of PPE equipment, particularly respirators, being procured from sources that do not meet the standards for domestically manufactured PPE. This should be done in full compliance with NIOSH, FDA, and OSHA standards. The Made in America Office should continuously update the product taxonomy and the critical components list. The reporting obligation should apply regardless of the legal mechanism authorizing nondomestic procurement, including the TAA coverage, BAA waivers, emergency authorities, micropurchases, simplified acquisitions, federal supply schedules, interagency acquisitions, and Berry and Kissell amendment determinations. Federal agencies should submit reports to a centralized database managed by the Made in America Office, which is updated routinely.

Thirty days before awarding a proposed nondomestic procurement contract exceeding a certain threshold, an agency should be required to publish a notice allowing domestic manufacturers to demonstrate capability. No waiver or nonavailability determination should remain in effect for more than 12 months when domestic capacity exists but is underused, 24 months when capacity is actively being developed, or 36 months when an entirely new capability must be built. Renewal would require demonstrating progress and Made in America Office approval. Any category under continuous waiver for five years would trigger a formal interagency review and report to the President and Congress. The Made in America Office should submit quarterly reports to relevant congressional committees detailing nondomestic procurement by agency, category, country, and mechanism and should publish an annual domestic capacity and supply chain vulnerability assessment.

Require Independent Audits of PPE Waiver Patterns and Nonavailability Determinations and Establish Enforceable Consequences for Agencies That Do Not Meet These Standards

Federal agencies that are procuring PPE should be subject to continuous monitoring and periodic audits of waiver patterns, nonavailability determinations, and domestic versus nondomestic sourcing rates, which the Made in America Office should conduct in coordination with agency inspectors general and GAO. Each nonavailability claim should be supported by market research conducted within a set period of solicitation, documented outreach to all known domestic manufacturers, analysis demonstrating that requirements do not exclude domestic products, and price comparison following a TCO analysis. The Made in America Office should maintain a centralized PPE procurement database tracking product category, domestic content percentage, country of manufacture, waiver justification, and manufacturer identity across all covered agencies and should publish routine reports to Congress on governmentwide PPE domestic sourcing rates, cross-agency waiver pattern discrepancies, and domestic industrial base conditions.

Agencies with persistently high waiver rates despite available domestic capacity should be required to submit corrective action plans with measurable milestones. Nonavailability determinations found in the audit to be inadequately supported, based on outdated market research, or inconsistent with other agencies' successful domestic procurement of the same items should be revoked and not renewed without new documented evidence of nonavailability. In addition, suppliers who provide wrong or incomplete information should be held accountable, and their contracts should be penalized.

Enabling Regulatory Compliance

A lack of established standards and certification requirements hindered new entrants to respirator manufacture. Although not generally seen as overly costly, the certification process can be challenging to navigate and has been complicated by regulatory body instability, notably including the recent disruption to NIOSH's PPE certification activity. Insufficient regulatory mechanisms also allowed counterfeit and ineffective PPE to flood the market, creating a fraught and noisy market environment.

Form Collaborative SME Networks for Compliance Resources

Smaller manufacturers should form collaborative networks to develop and share standardized compliance resources, including quality control procedure checklists, SOP templates, and equipment validation protocols for NIOSH-compliant production. This collective approach would lower barriers to certification, strengthen product legitimacy, and help smaller companies bring compliant PPE to market faster and at lower cost. Our interviewees noted that, in addition, smaller companies may also benefit from joining an industry association, such as the American Mask Manufacturers Association or the International Safety Equipment Association, to help maintain situational awareness of what is developing in the industry.²⁴ These organizations have helped enable other smaller companies to have their voices heard in the bigger PPE ecosystem.

Develop and Publish Accessible PPE Compliance Guidance

NIOSH and/or the FDA should develop and publish clear, accessible compliance guidance, including step-by-step certification checklists, equipment validation benchmarks, and SOP frameworks. These checklists could specifically be designed for small and emerging PPE manufacturers, reducing the knowledge gap that makes the certification process disproportionately burdensome for smaller firms.

²⁴ See also OSHA, “International Safety Equipment Association (ISEA),” webpage, undated-a.

Conclusion

The COVID-19 pandemic exposed a stark truth: The U.S. ability to protect health care workers and the public during a respiratory disease emergency is only as strong as its most fragile supply chain link. SMEs stepped forward to fill critical gaps during the pandemic, yet many have since exited the market, not because they could not produce quality products but largely because of the structural conditions of the U.S. PPE ecosystem. Without deliberate intervention, the United States may face the next biological outbreak with a domestic manufacturing base as depleted as, or more than, it was in 2020.

The central insight of this report is that domestic PPE resilience is not a problem of individual manufacturer capability but one of systemic design. The barriers SMEs face, such as prohibitive capital costs, material bottlenecks, fragmented certification, anticompetitive procurement, volatile demand, workforce erosion, and overcentralized stockpiling, are interconnected. Financing for automation means little if GPO practices lock SMEs out of hospital systems. Certification success means little without stable demand to retain a skilled workforce. A federal stockpile, no matter its size, is no substitute for an active manufacturing base ready to surge production on demand.

Addressing this challenge requires coordinated action across three mutually reinforcing fronts: lowering the barriers to SME entry and sustained operation, restructuring market access and demand signals, and redistributing preparedness responsibility across the federal and state levels.

Existing statutes—the Make PPE in America Act, the Berry and Kissell amendments, and the BAA—already articulate a national commitment to domestic sourcing. But commitments without enforcement, transparency, and structural support produce only the illusion of preparedness. Closing waiver loopholes, mandating procurement transparency, and tying compliance to meaningful consequences will determine whether these statutes achieve their intent.

Although all recommendations presented address documented barriers to SME surge capacity, demand-side interventions, particularly long-term federal contracts, hospital sourcing mandates, and stockpiling reforms, are likely to carry the greatest near-term impact, as SME willingness to invest in workforce and capital remains contingent on credible, sustained demand signals. On the supply side, warm-basing and expanded technical assistance offer the most durable returns by preserving the manufacturing readiness that erodes between crises. Together, these interventions form the highest-priority foundation from which the remaining regulatory and market fairness reforms can build.

These priorities translate into coordinated action across four areas:

- **Strengthening SME and domestic PPE manufacturing:** Congress and federal policymakers should consider pursuing a coordinated national strategy that treats SME resilience as integral to preparedness, supply chain security, and domestic industrial capacity. The broader preparedness and policy community should likewise align its support for PPE SMEs with its support for domestic manufacturing, recognizing that the two goals reinforce one another.
- **Overcoming supply, production, and capacity challenges:** Congress, the SBA, and relevant federal agencies should consider easing structural constraints by expanding affordable equipment financing,

supporting warm-basing and precrisis production readiness, funding technical assistance for automation and scale-up, and creating incentives to recruit and retain skilled manufacturing, quality assurance, and regulatory personnel.

- **Overcoming demand and market access constraints:** DHHS, DHS, DOW, the VA, CMS, ASPR, and state governments should create more-stable and predictable demand through long-term contracts, reformed stockpiling strategies, and SME sourcing requirements. Congress, the U.S. Trade Representative, the U.S. Department of Commerce, DHHS, the Internal Revenue Service, CMS, and state legislatures should consider improving SME price competitiveness through targeted tax incentives and trade tools. Additionally, CMS, FTC, the U.S. Department of Justice, Congress, the VA, and other oversight bodies should consider promoting fairness by curbing anticompetitive GPO and distributor practices.
- **Overcoming regulatory hurdles:** Congress, the FAR Council, GAO, agency inspectors general, and the Made in America Office should consider closing domestic-sourcing loopholes by tightening waiver criteria, requiring TCO analysis, and standardizing the definition of *domestically manufactured PPE*. NIOSH, the FDA, and SMEs themselves should lower compliance barriers through clearer certification guidance, standardized checklists, and collaborative SME networks, while recognizing that NIOSH's own resource constraints must be addressed for the approval process to have sustained improvement.

The window for action is narrowing. Each year of inaction erodes the manufacturing base further as SMEs exit the market and as the skill pipeline contracts. The next respiratory pandemic is not a question of if, but when. Whether the United States meets it with a diversified, resilient, domestically anchored PPE base or with a supply chain as fragile as the one exposed in 2020 depends on the choices stakeholders make now.

Preparedness is not a stockpile. It is an ecosystem. Building that ecosystem, and ensuring SMEs have a durable place within it, is one of the defining supply chain security challenges of the coming decade.

Methodology

Baseline Analysis of the Carnegie Mellon University Study

As a starting point, we evaluated the CMU analysis of barriers affecting SMEs during COVID-19 and identified research limitations that further analysis could address. The CMU analysis did not include EHMRs and examined only the 2020–2022 time frame, and we identified further research opportunities to assess whether the barriers still persisted in 2026, additional roadblocks that might have emerged affecting PPE SMEs, and the current status of the U.S. PPE industry.

Literature Review

Our literature review included both open-source media and peer-reviewed articles, focusing on key terms, including *PPE*, *mask manufacturing*, *COVID supply chains*, and *U.S. manufacturing*. We completed the literature review by pulling from databases using such research platforms as OpenAlex, ProQuest, Google Search, and large language models. This helped us review a range of sources, including research articles, books, dissertations, preprints, newspapers, and magazines. Our research inputs also included grey literature, such as government reports, technical reports, and company websites.

Semistructured Interviews

We interviewed several subject-matter experts and individuals experienced in supply networks, including those with experience in mask development and manufacturing, OSHA and NIOSH standards, emergency operations, public health responses, domestic manufacturing policy, and the SNS. We conducted semistructured interviews with respondents between January and early April 2026. Each interview lasted 30 to 60 minutes and was conducted in English. We created a guide to support the semistructured interviews, while still allowing participants the flexibility to expand on the questions based on their own experience and expertise. Each participant was informed of the note-taking and that their recorded responses would remain nonattributable. Prior to each interview, we provided the interviewee with a general overview of the project, previously identified barriers to SME market entry, and a summary of what we planned to discuss. These topics included

- experiences in manufacturing of respiratory PPE prior to, during, and post COVID-19 pandemic
- the role of several existing policies to drive domestic production and purchasing for PPE within the United States
- feedback on identified barriers to market entry, whether they were still in place, and challenges that were not previously noted
- regulatory processes for approval
- opportunities to support SMEs within the United States.

Limitations

We faced some limitations in identifying existing challenges for SMEs specializing in respiratory PPE to enter and successfully stay in the marketplace. First, following the COVID-19 pandemic, a significant number of SMEs closed or halted production of PPE, which limited our capacity to identify and interview experts. Many people were reluctant to talk, especially those in or with previous experience in federal agencies. Additionally, few manufacturers could speak to producing both N95 respirators and EHMRs, with the majority of interviewees speaking to their experiences in N95 and disposable mask production. This suggests that SMEs were underrepresented in the sample; although we sought to broaden the scope beyond the list provided by Nikhil et al., future research would be strengthened by incorporating a state-by-state analysis.

Focus on U.S. Respiratory PPE Production by SMEs

Finally, our research focused on domestic respiratory PPE production in SMEs, which limited our scope in understanding what large and foreign enterprises experienced during pandemic surge manufacturing. Additionally, our research and interviews revealed that barriers to both market entry and sustained participation extend well beyond those previously identified. These challenges are deeply embedded in the broader political economy of domestic PPE purchasing and production, where the landscape is largely shaped by dominant large firms and hospital purchasing groups. Moreover, there is evidence of strategic behavior aimed at circumventing policy enforcement, further entrenching existing advantages and limiting opportunities for new or smaller entrants.

COVID-19–Era Policy Measures

Federal Measures

Several COVID-19–era laws provided funding to respond to the pandemic. These funding options usually had very broad funding purposes, so domestic manufacturing of PPE was eligible for funding but competed with other activities.

Public Law 116-123, Coronavirus Preparedness and Response Supplemental Appropriations Act, March 6, 2020

This law provided \$61 million to respond to COVID-19, including “advanced manufacturing for medical products, the monitoring of medical product supply chains, and related administrative activities.” The law also provided another \$3.1 billion to respond to COVID-19,

including the development of necessary countermeasures and vaccines, prioritizing platform-based technologies with U.S.-based manufacturing capabilities, and the purchase of vaccines, therapeutics, diagnostics, necessary medical supplies, medical surge capacity, and related administrative activities.

Public Law 116-139, Paycheck Protection Program and Health Care Enhancement Act, April 24, 2020

This law provided \$25 billion for the Public Health and Social Services Emergency Fund within DHHS, which included “the manufacturing, procurement and distribution of tests, testing equipment and testing supplies, including personal protective equipment needed for administering tests.”

Public Law 116-136, Coronavirus Aid, Relief, and Economic Security Act, March 27, 2020

The Coronavirus Aid, Relief, and Economic Security (CARES) Act provided

\$80,000,000, to remain available until expended, to prevent, prepare for, and respond to coronavirus, domestically or internationally, including funds for the development of necessary medical countermeasures and vaccines, advanced manufacturing for medical products, the monitoring of medical product supply chains, and related administrative activities.¹

¹ Pub. L. 116-136, 2020.

The law further declared that

\$50,000,000 shall be for the Hollings Manufacturing Extension Partnership to assist manufacturers to prevent, prepare for, and respond to coronavirus and \$10,000,000 shall be for the National Network for Manufacturing Innovation (also known as “Manufacturing USA”) to prevent, prepare for, and respond to coronavirus, including to support development and manufacturing of medical countermeasures and bio-medical equipment and supplies.²

Some of this funding went to help manufacturers pivot to COVID-related manufacturing, including the production of PPE, and created portals to connect purchasers with suppliers.³ The law also specified that

an additional amount for “Public Health and Social Services Emergency Fund,” \$27,014,500,000, . . . remain available until September 30, 2024, to prevent, prepare for, and respond to coronavirus, domestically or internationally, including the development of necessary countermeasures and vaccines, prioritizing platform-based technologies with U.S.-based manufacturing capabilities, the purchase of vaccines, therapeutics, diagnostics, necessary medical supplies, as well as medical surge capacity, addressing blood supply chain, workforce modernization, telehealth access and infrastructure, initial advanced manufacturing, novel dispensing, enhancements to the U.S. Commissioned Corps, and other preparedness and response activities: Provided, That funds appropriated under this paragraph in this Act may be used to develop and demonstrate innovations and enhancements to manufacturing platforms to support such capabilities.⁴

Some of this funding went to companies building domestic PPE manufacturing capacity.⁵ However, at least some of the companies funded to build manufacturing capacity have struggled to manufacture PPE.⁶

State Measures

It is important to note that some of the funding in this category may have stemmed from federal COVID-19-era funding, such as the CARES Act, which was sometimes distributed through states. In addition, this funding might have been used to temporarily pivot to manufacturing PPE, instead of permanently enhancing domestic PPE production capacity. Georgia provided \$1,500 in tax credits for qualifying jobs engaged in PPE manufacturing within the state, which expired January 1, 2025. Georgia House Bill 846, signed into law June 30, 2020, provided \$1,250 in tax credits for qualifying jobs engaged in PPE manufacturing. Massachusetts distributed \$9.5 million in grants to manufacturers to produce PPE. Maryland created a \$5 million fund with grants of up to \$100,000 for manufacturers to pivot to PPE production or to enhance existing production capacity. Michigan provided grants of up to \$150,000 for small manufacturers to produce PPE. Virginia provided funding through its Virginia Jobs Investment Program to at least one PPE manufacturer to help expand operations. New York created the Production & Development of Medical/Critical Need Supplies Related to Covid 19 Financial Assistance Program, which provided grants to existing organizations to pivot to or expand PPE production.

² Pub. L. 116–136, 2020.

³ NIST, “Impacts of CARES Act Funding Across the MEP National Network,” February 2022.

⁴ Pub. L. 116–136, 2020.

⁵ U.S. Department of Defense, “DOD Announces \$226.3 Million in Industrial Base Expansion Actions,” press release, May 28, 2021.

⁶ Nell Greenfieldboyce, “More Medical Gloves Are Coming from China, as U.S. Makers of Protective Gear Struggle,” NPR, November 3, 2023.

Executive Orders

Two EOs sought to strengthen the U.S. capacity to supply PPE in response to public health emergencies.

EO 13944, Combating Public Health Emergencies and Strengthening National Security by Ensuring Essential Medicines, Medical Countermeasures, and Critical Inputs Are Made in the United States, August 6, 2020

This EO stated that it was the policy of the United States to

accelerate the development of cost-effective and efficient domestic production of Essential Medicines and Medical Countermeasures [including PPE] and have adequate redundancy built into the domestic supply chain for Essential Medicines, Medical Countermeasures, and Critical Inputs.

The EO required the U.S. Trade Representative to modify federal procurement covered under free trade agreements “to exclude coverage of Essential Medicines, Medical Countermeasures, and Critical Inputs.” It directed the DHHS secretary to identify and mitigate vulnerabilities in the supply chain of medical countermeasures.

EO 14001, A Sustainable Public Health Supply Chain, January 21, 2021

In addition to focusing on immediate PPE needs, the EO required the heads of several executive agencies to create “a strategy to design, build, and sustain a long-term capability in the United States to manufacture supplies for future pandemics and biological threats.” The strategy must include “an analysis of the role of foreign supply chains in America’s pandemic supply chain, America’s role in the international public health supply chain, and options for strengthening and better coordinating global supply chain systems in future pandemics.”

Types of Respirators

Although a variety of PPE types are used to protect workers, we focused on respiratory protection because the COVID-19 pandemic (the result of a respiratory pathogen) served as a critical inflection point for U.S. domestic supply chain resilience. The evolution of PPE over the years has focused on ensuring the highest level of protection and ease and comfort of use and on increasing attention to reuse and environmental waste. Standards now exist that explicitly state requirements for PPE, such as those for the fluid resistance levels for gowns set by the Association for the Advancement of Medical Instrumentation and approval of respiratory protection by the NIOSH.¹ Respiratory PPE, however, primarily falls into two categories—APRs and atmosphere-supplying respirators—which determine the type of protection needed. This is further broken down into five NIOSH-approved kinds of APRs and atmosphere-supplying respirators for respiratory protection against chemical, biological, radiological, and nuclear agents: elastomeric half facepiece respirators, elastomeric full facepiece respirators, filtering facepiece respirators (FFRs), powered APRs, and supplied-air respirators (self-contained breathing apparatus), as well as combination respirators. We focused on APRs because they were heavily used during the COVID-19 pandemic but also because they represent the most likely respiratory protection for general uptake in future biological events.

APRs and masks work by filtering out the particles in the air and blocking droplets from inhalation. Through this, they offer varied levels of protection to the wearer against respiratory hazards, including infectious aerosols and droplets. APRs are the most commonly used respiratory PPE in health care, in public health, and by the public and include FFRs, elastomeric respirators, and powered APRs. These masks rely on filters, cartridges, or canisters to remove vapors and/or aerosols from the air, allowing the wearer to breathe in filtered air and reduce the risk of inhaling infectious particles. The effectiveness of these respirators depends heavily on proper fit and use, meaning that the wearer should be appropriately fit-tested and should ensure that they are adhering to manufacturers' recommendations for use. During the COVID-19 pandemic, FFRs were the most common masks, offering varied levels of filtration, with barrier face coverings, disposable masks, and cloth masks offering varied levels of filtration.² Of those, FFRs, such as N95s, provide the highest level of infectious particle protection for disposable masks but do not protect the wearer against gases or vapors. There are nine filter classes for FFRs, falling into three categories of filtration efficiency (see Table C.1).³

To ensure quality, those NIOSH-approved FFRs continue to meet NIOSH requirements, meaning that they have been manufactured and tested to meet these standards. During the COVID-19 pandemic, as supply chains were severely affected, other FFRs were used, referred to as International FFRs, such as KN95s, which

¹ CDC, "NIOSH Personal Protective Equipment Information (PPE-Info): ANSI/AAMI PB70—Class 3," webpage, last reviewed February 21, 2024; NIOSH, "Respirator Types and Use," webpage, Centers for Disease Control and Prevention, March 4, 2025e.

² NIOSH, "Respirators and Mask Types and Performance," webpage, Centers for Disease Control and Prevention, March 25, 2025f.

³ NIOSH, "Filtering Facepiece Respirators," webpage, Centers for Disease Control and Prevention, February 3, 2025c.

TABLE C.1
Filtering Facepiece Respirator Classes

Class	Filter	Ability to Filter Airborne Particles	Resistance to Oil
N	N95	At least 95 percent	Not resistant
	N99	At least 99 percent	Not resistant
	N100	At least 99.97 percent	Not resistant
R	R95	At least 95 percent	Somewhat resistant
	R99	At least 99 percent	Somewhat resistant
	R100	At least 99.97 percent	Somewhat resistant
P	P95	At least 95 percent	Strongly resistant
	P99	At least 99 percent	Strongly resistant
	P100	At least 99.97 percent	Strongly resistant

were imported to the United States. Per NIOSH, “International FFRs are a disposable type of respirator that achieve a level of filtration efficiency based on the specified standard with a minimum of 80%.”⁴ During this time, the efficacy and multiuse capabilities of EHMRs were realized, which popularized their use in health care and by the public as a means of protecting against SARS-CoV-2/COVID-19.

EHMRs protect against gases, vapors, and particles. Unlike FFRs, such as N95s—which are constructed from layers of nonwoven polypropylene fiber with rubber straps and a metal nose bar—EHMRs are built from silicone, polymeric, or synthetic or natural rubber materials, with detachable filters, making them cleanable with disinfection and reusable. Many models also feature exhalation valves that reduce breathing resistance, although, when present, these valves should include a filter to prevent the spread of infectious particles from a contagious wearer. Ultimately, the introduction of EHMRs into public health and health care response during COVID-19 provided safer and sustainable high-level respiratory protection.

In sum, the most protective masks for highly infectious respiratory pathogens, such as SARS-CoV-2/COVID-19 and measles, are those of higher filtration, such as N95s and EHMRs, all of which require fit-testing to ensure a tight seal against the face. Although disposable and cloth masks were used significantly during the COVID-19 pandemic, their efficacy was highly variable. Despite the range of respiratory protection available, our analysis primarily focused on N95 and EHMR respirators but references previous research and production regarding disposable masks accordingly.

⁴ NIOSH, 2025f.

Case Study

PPE supply chains and access issues are not unique to the COVID-19 pandemic. Several other public health emergencies have highlighted significant stressors for U.S. hospitals and preparedness efforts, including the 2009 H1N1 outbreak and the 2014–2016 Ebola virus disease outbreak. In 2009, a novel influenza virus emerged, which was distinctly different and officially designated as the influenza A (H1N1) pdm09 virus. The H1N1 influenza pandemic spread rapidly after early identification in North America, with the World Health Organization declaring a pandemic in June 2009, affecting 74 countries. By May 3, 2009, the CDC had deployed 25 percent of the SNS to aid states in response. This flu primarily affected children and middle-aged adults. The CDC estimated that there were more than 60 million cases between April 12, 2009, and April 10, 2010, with roughly 12,469 deaths in the United States. Worldwide, there were between 151,700 and 575,400 estimated deaths.¹ In this novel situation, lack of exact information on transmission, virulence, contagiousness, and recommendations for PPE and infection prevention caused significant confusion among health care workers and responders. With changing guidance, there were reports of PPE shortages, with health departments taking measures to provide additional N95 respirators for health care facilities.² In addition to supply issues, several infection preventionists reported obtaining incorrect supplies from regional or national stockpiles, not knowing which types of respiratory protection to use when caring for novel H1N1 patients, and attempting to fit-test large numbers of staff during short, emergent periods led to logistical issues.³

In 2014, the United States faced a similar situation, in which changing PPE guidance and a health care system scrambling to respond translated to significant PPE supply chain hurdles. In an unprecedented outbreak of Ebola virus disease, more than 28,610 cases occurred across Guinea, Liberia, and Sierra Leone. During this time, the United States treated 11 people for Ebola virus disease, two of whom died. The majority were infected while caring for patients outside the United States and medically evacuated; however, four cases were identified domestically. Of these four, two cases were infected outside the United States, and two were nurses who cared for an Ebola patient and contracted the disease during the course of care in Dallas, Texas. The identification of a patient with Ebola virus disease in Dallas and nosocomial transmission ignited a panic across U.S. hospitals as they desperately attempted to prepare for potential patients. The PPE requirements for safely treating a patient with Ebola are substantial, requiring not just respiratory protection but also gowns, several layers of gloves, boot covers, face protection, and designated donning and doffing areas. Seventy-one percent of hospital administrators reported that their hospitals were not prepared to receive an Ebola patient, and with rapidly evolving infection prevention and patient care guidance from the CDC,

¹ CDC, “2009 H1N1 Pandemic (H1N1pdm09 virus),” webpage, last reviewed June 11, 2019.

² “Policy Research and Implementation: Healthcare Systems, Standards, and Certification,” Chapter 5 in Elaine L. Larson and Catharyn T. Liverman, eds., *Preventing Transmission of Pandemic Influenza and Other Viral Respiratory Diseases: Personal Protective Equipment for Healthcare Personnel, Update 2010*, Institute of Medicine, National Academies Press, 2011.

³ Terri Rebmann and William Wagner, “Infection Preventionists’ Experience During the First Months of the 2009 Novel H1N1 Influenza A Pandemic,” *American Journal of Infection Control*, Vol. 37, No. 10, December 2009.

efforts to purchase and maintain adequate PPE stockpiles were stressed.⁴ During this time, hospitals scrambled to maintain up-to-date PPE supplies, as well as staff competency, with overall cost for acute-care hospitals to prepare for Ebola across the United States estimated to be around \$360 million.⁵

In both public health emergencies—the 2009 H1N1 pandemic and 2014 Ebola outbreak—the response in the United States was affected by confusion about infection prevention guidance, PPE recommendations and supply chains, and a health care infrastructure that struggled to maintain readiness. Each time, the commercial supply chain of PPE had very little flexibility and agility to respond to public health emergencies and corresponding large, unexpected increases in demand. These gaps and the reliance on foreign suppliers were highlighted significantly in the years prior to the COVID-19 pandemic.⁶

⁴ Suzanne Murrin, *Hospitals Reported Improved Preparedness for Emerging Infectious Diseases After the Ebola Outbreak*, Office of Inspector General, U.S. Department of Health and Human Services, OEI-06-15-00230, October 2018.

⁵ Michael A. Smit, Kenneth A. Rasinski, Barbara I. Braun, Linda L. Kusek, Aaron M. Milstone, Daniel J. Morgan, and Leonard A. Mermel, “Ebola Preparedness Resources for Acute-Care Hospitals in the United States: A Cross-Sectional Study of Costs, Benefits, and Challenges,” *Infection Control and Hospital Epidemiology*, Vol. 38, No. 4, April 2017.

⁶ Anita Patel, Maryann M. D’Alessandro, Karen J. Ireland, W. Greg Burel, Elaine B. Wencil, and Sonja A. Rasmussen, “Personal Protective Equipment Supply Chain: Lessons Learned from Recent Public Health Emergency Responses,” *Health Security*, Vol. 15, No. 3, June 2017; Saskia Popescu and Rebecca Leach, “Identifying Gaps in Frontline Healthcare Facility High-Consequence Infectious Disease Preparedness,” *Health Security*, Vol. 17, No. 2, April 2019.

Baseline PPE Demand, by Critical Sector

The following list presents the PPE employment for each critical sector:

- health care—4,278,560
- research and development—460,224
- support services—313,434
- manufacturing—233,470
- chemical manufacturing—172,025
- utilities—153,567
- maintenance—150,318
- metal and materials manufacturing—138,646
- food manufacturing—125,931
- heavy construction—107,405
- network access—88,871
- transportation support—76,727
- insurance—72,133
- Global Positioning System and cyber—68,169
- education—51,630
- waste management—48,980.

Abbreviations

APR	air-purifying respirator
ASPR	Administration for Strategic Preparedness and Response
BAA	Buy American Act
CARES	Coronavirus Aid, Relief, and Economic Security
CDC	U.S. Centers for Disease Control and Prevention
CMS	Centers for Medicaid and Medicare Services
CMU	Carnegie Mellon University
COVID-19	coronavirus disease 2019
DHHS	U.S. Department of Health and Human Services
DHS	U.S. Department of Homeland Security
DOW	U.S. Department of War
DPA	Defense Production Act
EHMR	elastomeric half mask respirator
EO	Executive Order
FAR	Federal Acquisition Regulation
FDA	U.S. Food and Drug Administration
FEMA	Federal Emergency Management Agency
FFR	filtering facepiece respirator
FTC	Federal Trade Commission
GAO	U.S. Government Accountability Office
GPO	group purchasing organization
MEP	Manufacturing Extension Partnership
NIOSH	National Institute for Occupational Safety and Health
NIST	National Institute of Standards and Technology
OSHA	Occupational Safety and Health Administration
PPE	personal protective equipment
SARS-CoV-2	severe acute respiratory syndrome coronavirus 2
SBA	U.S. Small Business Administration
SLEP	Shelf-Life Extension Program
SME	small and medium-sized enterprises
SNS	Strategic National Stockpile
SOP	standard operating procedure
TAA	Trade Agreements Act
TCO	total cost of ownership
VA	U.S. Department of Veterans Affairs
VMI	vendor-managed inventory

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